



PAN3029 / PAN3060 系列

产品说明书

V1.7 Mar. 2024

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低功耗远距离无线收发芯片



Overview

PAN3029 / PAN3060 is a low-power long-distance wireless transceiver chip using ChirpIoT™ modulation and demodulation technology. It supports half-duplex wireless communication and operates in the frequency bands of 408~565MHz and 816~1080MHz. The chip has the characteristics of high anti-interference, high sensitivity, low power consumption and ultra-long transmission distance. It has a maximum sensitivity of -143dBm and a maximum output power of 20dBm, which produces an industry-leading link budget, making it the best choice for long-distance transmission and applications with extremely high reliability requirements. Compared with conventional modulation technology, PAN3029 / PAN3060 also has significant advantages in blocking and adjacent channel selection, which can further improve communication reliability. At the same time, it also provides greater flexibility. Users can adjust the spread spectrum modulation bandwidth, spreading factor and error correction rate by themselves, effectively improving the trade-off between distance, anti-interference ability and power consumption of chips using conventional modulation technology.

Key Features

- Frequency: 408~565MHz, 816~1080MHz
- Modulation: ChirpIoT™
- Transmit power: -30dBm~20dBm
- Maximum link budget: 163dB
- Sensitivity: -143dBm@62.5kHz
- Current consumption:
 - Sleep current: 200nA
 - Receiving current: 4.1mA@DCDC mode
 - Transmitting current:
 - 91mA@20dBm
 - 13mA@-2dBm
- Bandwidth
 - PAN3029: 62.5kHz、125kHz、250kHz、500kHz
 - PAN3060: 125kHz、250kHz、500kHz
- SF factor and automatic identification of spreading factor
 - PAN3029: 5~12
 - PAN3060: 5~9
- Coding rate: 4/5, 4/6, 4/7, 4/8
- CAD function
- Data rate: 0.08~59.9kbps
- 4-wire SPI/3-wire SPI/I2C configuration interface, 4 GPIOs

- Operating Voltage: 1.8V ~3.6V (DCDC mode 2V~3.6V)
- Working temperature: -40°C~85°C
- Package: QFN28 (4×4mm)

Typical Applications

- Smart Factory
- Smart Water
- Smart Agriculture
- Smart Healthcare
- Smart Community
- Smart Fire Protection



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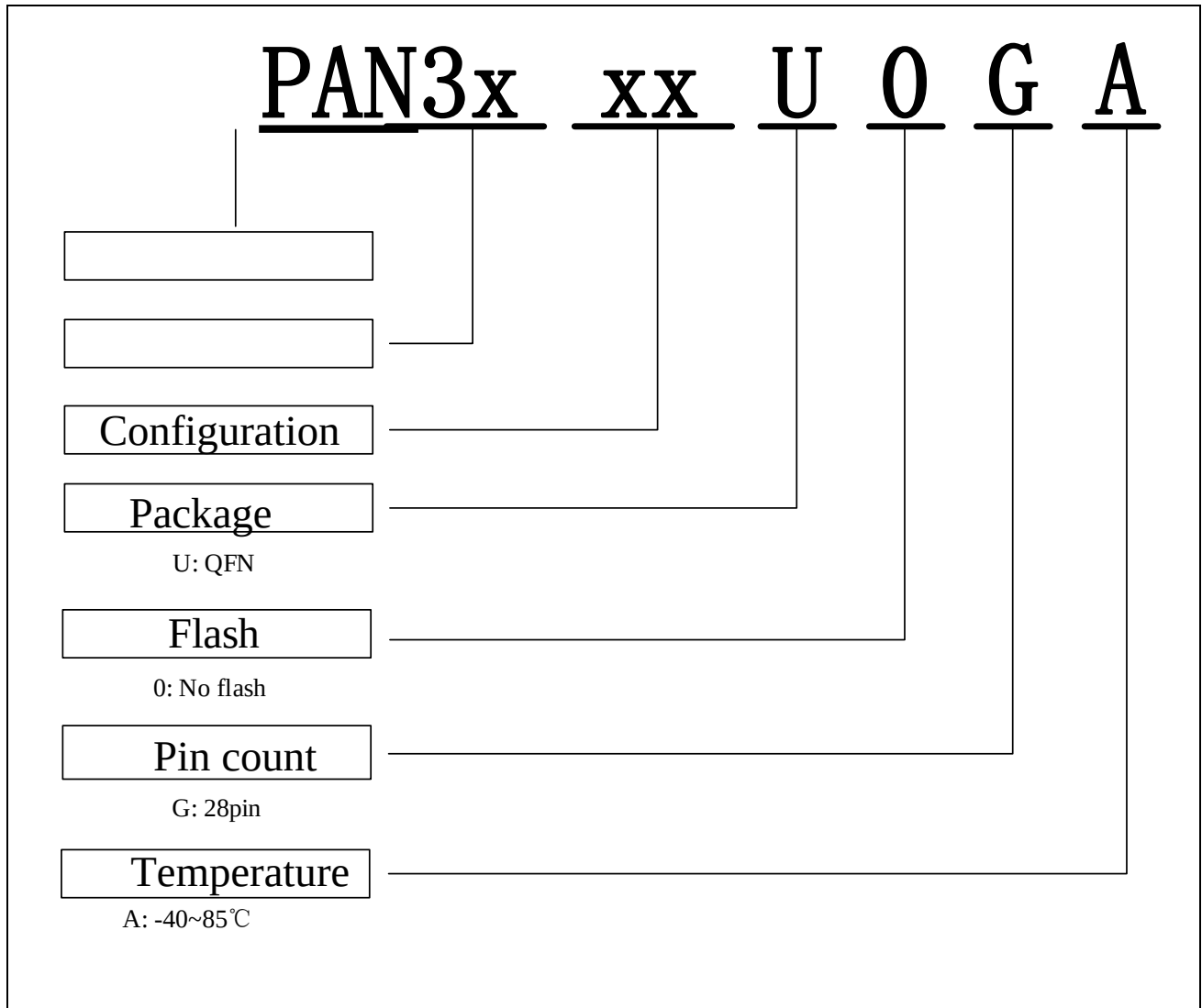
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1 Naming convention





2 Ordering Information

Model	Chip Type	Package	Pin count	IO	BW	SF	Bit rate	Temperature	Package
PAN3029 U0GA	Sub-1G	QFN	28	4	62.5kHz 125kHz 250kHz 500kHz	SF5~SF12	0.08~ 59.9kbps	-40~85°C	Tape & Reel
PAN3060 U0GA	Sub-1G	QFN	28	4	125kHz 250kHz 500kHz	SF5~SF9	0.5~ 59.9kbps	-40~85°C	Tape & Reel

Before ordering, please consult sales for the latest mass production information.

This manual describes the maximum package and the most complete functions. For the configuration differences of each model, please refer to the ordering information on this page.

3 System structure block diagram

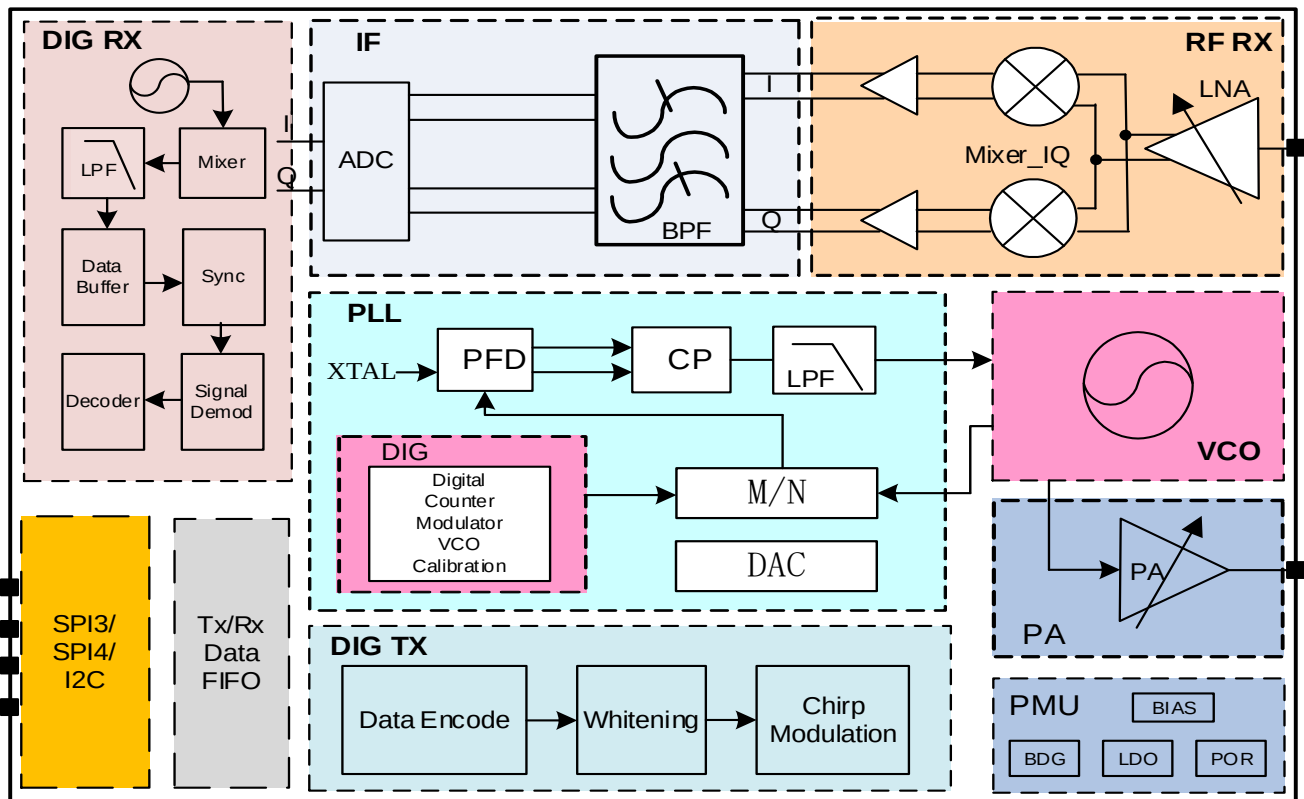


Figure 3-1 System structure block diagram

4 Pin Definition and Description

4.1 Pin Diagram

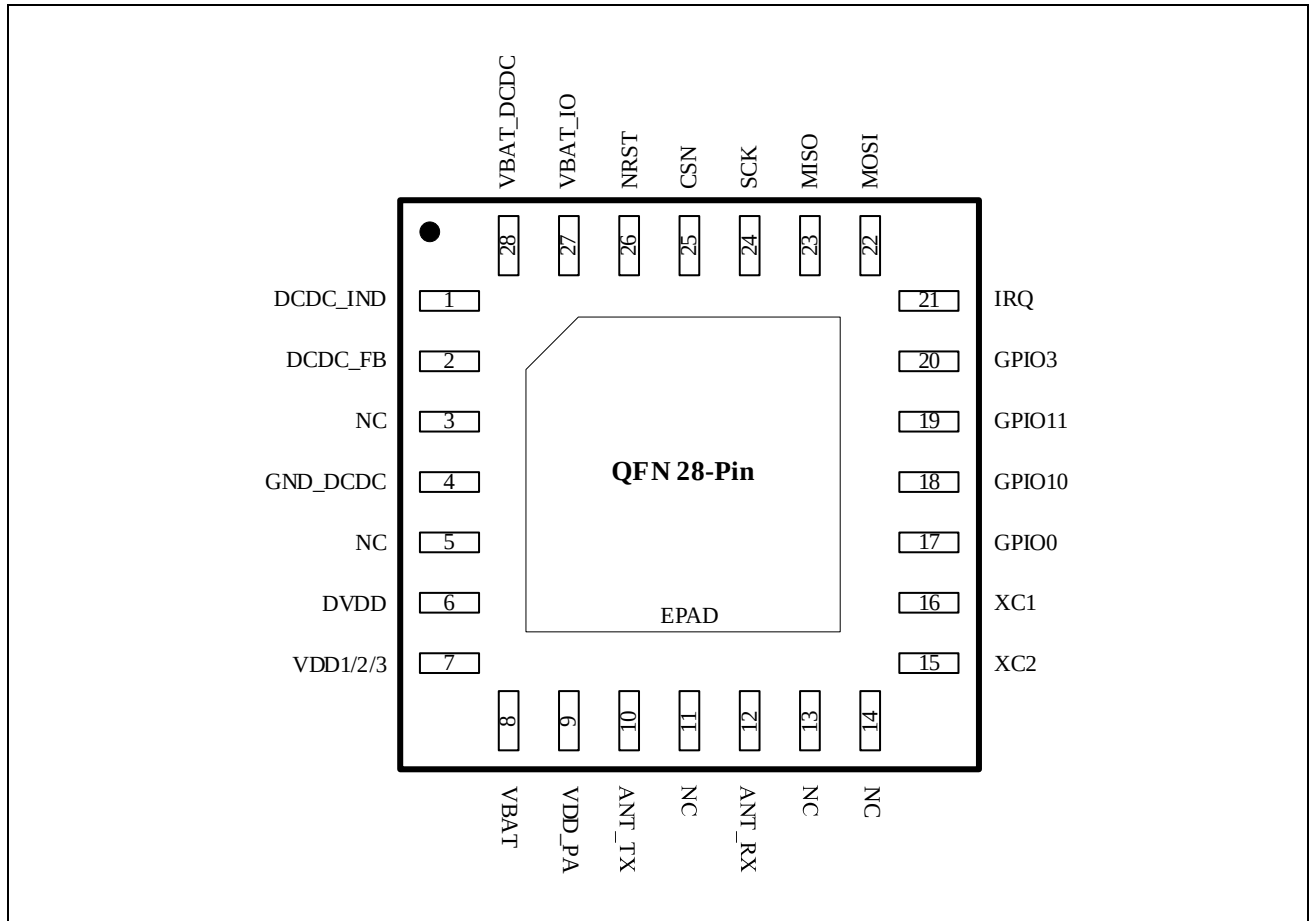


Figure 4-1 Pin diagram

4.2 Pin Description

Table 4-1 Pin Description

Pin No.	Pin Name	Type	Description
1	DCDC_IND	AO	内部 DCDC 输出，连接外部串联电感
2	DCDC_FB	AI	内部 DCDC 反馈输入，DCDC 模式与 VDD1/2/3 相连
3	NC		
4	GND_DCDC	G	模拟地
5	NC		
6	DVDD	P	数字电源 LDO 输出
7	VDD1/2/3	P	模拟电源，DCDC 模式连接 VFB，非 DCDC 模式连接总电源
8	VBAT	P	模拟电源，连接总电源
9	VDD_PA	P	PA 电源 LDO 输出
10	ANT_TX	AO	发射端 PA 输出
11	NC		
12	ANT_RX	AI	接收端 LNA 输入
13	NC		
14	NC		
15	XC2	AI	晶振输入
16	XC1	AO	晶振输出
17	GPIO0	I/O	数字 IO
18	GPIO10	I/O	数字 IO
		O	外置 PA 控制信号
19	GPIO11	I/O	数字 IO
		O	信道状态指示信号
20	GPIO3	I/O	数字 IO
21	IRQ	O	中断信号
22	MOSI	I	SPI 数据输入信号
23	MISO	O	SPI 数据输出信号
24	SCK	I	SPI 串行时钟
25	CSN	I	SPI 片选信号
26	NRST	I	复位
27	VBAT_IO	P	数字 GPIO 电源，连接总电源
28	VBAT_DCDC	P	DCDC 电源，连接总电源



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29	E-PAD	G	芯片底部 GND 焊盘，需要接地
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5 Electrical Characteristics

All parameters in this section are based on TA=25°C, power supply voltage 3.3V test, unless otherwise specified.

5.1 Absolute Maximum Ratings

Table 5-1 Absolute Maximum Ratings

Parameter	Description	Condition	Min	Typical	Max	Unit
VDD	VDD1/VDD2/VDD3/VBAT/VBAT_IO	Freq: 490MHz	-0.3	3.3	3.6	V
V _I	输入电压		-0.3	-	VDD	V
V _O	输出电压		VSS	-	VDD	V
T _{OP}	工作温度		-40	-	85	°C
T _{STG}	存储温度		-55	-	125	°C

Note: Exceeding one or more limit values may cause permanent damage to the chip. Electrostatically sensitive device, comply with protection rules when handling.

5.2 DC characteristics

Test conditions:

- Frequency: 490 MHz

Table 5-2 Voltage and current (frequency 490MHz)

Parameter	Description	Min	Typical	Max	Unit	Condition
VDD	电源	1.8	3.3	3.6	V	T _A =25°C, LDO 模式
		2	3.3	3.6	V	T _A =25°C, DCDC 模式
VSS	地	-	0	-	V	-
I _{DeepSleep}	深度睡眠电流	-	200	-	nA	-
I _{TX,20dBm}	TX 模式的工作电流	-	91	-	mA	20dBm 输出功率
I _{TX,18dBm}	TX 模式的工作电流	-	77	-	mA	18dBm 输出功率
I _{TX,-2dBm}	TX 模式的工作电流	-	13	-	mA	-2dBm 输出功率
I _{RX,LDO}	RX 模式的工作电流	-	7.8	-	mA	LDO 模式下, 最大 LNA 增益
I _{RX,DCDC}	RX 模式的工作电流	-	4.1	-	mA	DCDC 模式下, 最大 LNA 增益
V _{OH}	输出高电平电压	VDD-0.3	-	VDD	V	-
V _{OL}	输出低电平电压	VSS	-	VSS+0.3	V	-
V _{IH}	输入高电平电压	0.8*VDD	-	-	V	-

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V _{IL}	输入低电平电压	-	-	0.2*VDD	V	-
SPI_rate	SPI 速率	-	-	10	Mbps	-

Test conditions:

Frequency: 915MHz

Table 5-3 Voltage and current (frequency 915MHz)

Parameter	Description	Min	Typical	Max	Unit	Condition
VDD	电源	1.8	3.3	3.6	V	T _A =25°C, LDO 模式
		2	3.3	3.6	V	T _A =25°C, DCDC 模式
VSS	地	-	0	-	V	-
I _{DeepSleep}	深度睡眠电流	-	200	-	nA	-
I _{TX,20dBm}	TX 模式的工作电流	-	100	-	mA	20dBm 输出功率
I _{TX,18dBm}	TX 模式的工作电流	-	81	-	mA	18dBm 输出功率
I _{TX,-2dBm}	TX 模式的工作电流	-	15	-	mA	-2dBm 输出功率
I _{RX,LDO}	RX 模式的工作电流	-	7.8	-	mA	LDO 模式下, 最大 LNA 增益
I _{RX,DCDC}	RX 模式的工作电流	-	4.1	-	mA	DCDC 模式下, 最大 LNA 增益
V _{OH}	输出高电平电压	VDD-0.3	-	VDD	V	-
V _{OL}	输出低电平电压	VSS	-	VSS+0.3	V	-
V _{IH}	输入高电平电压	0.8*VDD	-	-	V	-
V _{IL}	输入低电平电压	-	-	0.2*VDD	V	-
SPI_rate	SPI 速率	-	-	10	Mbps	-

5.3 RF Characteristics

Test conditions:

- Frequency: 490MHz
- Error Correction Code = 4/8
- Packet error rate $\leq 5\%$
- Payload length = 10 Bytes

Table 5-4 General RF characteristics (frequency 490MHz)

Parameter	Description	Condition	Min.	Typical	Max.	Unit
通用频率						
F _{op}	工作频率	-	408	-	565	MHz
		-	816	-	1080	MHz
F _{xtal}	晶振频率	-	-	32	-	MHz
R _S	晶体串联电阻	-	-	30	50	Ω
C _{FOOT}	晶体外部电容	-	8	15	22	pF
C _{LOAD}	晶体负载电容	-	6	10	12	pF
F _{TOL}	初始频率容限	-	-	± 10	-	ppm
BR	比特速率 (PAN3029 系列)	-	0.08	-	59.9	kbps
	比特速率 (PAN3060 系列)	-	0.5	-	59.9	kbps
发射器						
P _{LPWAN}	输出功率	-	-30	-	20	dBm
接收器						
RF_62.5 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 62.5 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-129 -137 -143	- - -	dBm
RF_125 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 125 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-126 -134 -140	- - -	dBm
RF_250 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 250 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-124 -132 -137	- - -	dBm
RF_500 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 500 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-121 -129 -134	- - -	dBm
RF_62.5 (DCDC)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 62.5 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-125 -133 -139	- - -	dBm
RF_125 (DCDC)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 125 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-122 -130 -137	- - -	dBm
RF_250 (DCDC)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 250 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-120 -128 -134	- - -	dBm

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RF_500 (DCDC)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 500 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-117 -125 -130	- - -	dBm
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Test conditions:

- Frequency: 915MHz
- Error Correction Code = 4/8
- Packet error rate $\leq$ 5%
- Payload length = 10 Bytes

Table 5-5 General RF characteristics (frequency 915MHz)

Parameter	Description	Condition	Min.	Typical	Max.	Unit
通用频率						
F _{op}	工作频率	-	408	-	565	MHz
		-	816	-	1080	MHz
F _{xtal}	晶振频率	-	-	32	-	MHz
R _S	晶体串联电阻	-	-	30	50	Ω
C _{FOOT}	晶体外部电容	-	8	15	22	pF
C _{LOAD}	晶体负载电容	-	6	10	12	pF
F _{TOL}	初始频率容限	-	-	± 10	-	ppm
BR	比特速率 (PAN3029 系列)	-	0.08	-	59.9	kbps
	比特速率 (PAN3060 系列)	-	0.5	-	59.9	kbps
发射器						
P _{LPWAN}	输出功率	-	-30	-	20	dBm
接收器						
RF_62.5 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 62.5 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-128 -136 -141	- - -	dBm
RF_125 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 125 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-125 -132 -138	- - -	dBm
RF_250 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 250 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-121 -129 -135	- - -	dBm
RF_500 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 500 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-119 -127 -132	- - -	dBm
RF_62.5 (DCDC)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 62.5 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-128 -136 -140	- - -	dBm
RF_125 (DCDC)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 125 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-125 -132 -137	- - -	dBm
RF_250	RF 灵敏度, 长距离模式, 最高 LNA 增益,	SF = 7	-	-120	-	dBm



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(DCDC)	使用分离的 RX/TX 通道 250 kHz 带宽	SF = 10	-	-129	-	
		SF = 12	-	-135	-	
RF_500 (DCDC)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 500 kHz 带宽	SF = 7	-	-118	-	dBm
		SF = 10	-	-126	-	
		SF = 12	-	-130	-	

6 Frequency synthesizer

The chip integrates high-performance RF transceiver circuits and can support multiple Sub-1GHz bands such as 400M, 800M, etc.

The chip contains a high-performance, low phase noise, fully integrated fractional frequency synthesizer. The frequency synthesizer is used to generate the LO frequency used by the down-conversion mixer in the receive mode. In the transmit mode, it is also used to directly generate the modulated RF carrier. The fractional frequency synthesizer provides excellent phase noise performance, frequency resolution less than 100Hz, and has low power consumption. The frequency synthesizer has a fast frequency stabilization function, which can complete the receive and transmit frequency stabilization in a very short time to optimize the chip power consumption.

7 Modem

The chip uses a modem that supports ChirpIoT™ transmission, and achieves an ultra-high communication link budget through excellent spread spectrum modulation technology and forward error correction technology. Compared with the traditional FSK-based modulation method, the range and stability of the radio communication link are increased.

The ChirpIoT™ modem has greatly enhanced its anti-interference ability. Compared with the FSK modulation method, the ChirpIoT™ modem can suppress co-channel interference by up to 19dB, that is, the interference signal energy can be 19dB higher than the ChirpIoT™ signal, which greatly ensures that the ChirpIoT™ signal can simply coexist with other modulation systems and is widely used in hybrid network communications.

7.1 Modulation parameters

The chip optimizes the transmission performance of ChirpIoT™ signals by configuring different modulation parameters to achieve a balance between transmission rate, transmission distance, and transmission reliability. The specific parameters are as follows:

- Signal bandwidth (BW)
- Spreading Factor (SF)
- Coding rate (CR)
- Low rate mode (LDR)

7.1.1 Signal bandwidth

PAN3029 supports four signal bandwidths: 500kHz, 250kHz, 125kHz, and 62.5kHz. PAN3060 supports three signal bandwidths: 500kHz, 250kHz, and 125kHz.

Different signal bandwidths can be selected through register configuration.

7.1.2 Spreading Factor

PAN3029 can support 8 different spreading factors from SF5 to SF12, and PAN3060 can support 5 different spreading factors from SF5 to SF9. The larger the spreading factor, the lower the data transmission payload rate.

7.1.3 Coding Rate

The chip uses forward error correction technology and supports four different coding rates: CR45, CR46, CR47, and CR48. This forward error correction technology can correct error bits within a certain range and improve transmission reliability.

7.1.4 Low rate mode

The low-rate mode is a communication mode of ChirpIoT™ modulation technology, which improves the reliability of signal transmission by reducing the effective bit information carried per unit time.

7.2 Frame structure

The frame structure of ChirpIoT™ transmission signal consists of four parts: signal Preamble, Header, Payload, and CRC, as shown in the following figure.



Figure 7-1 PHY frame structure

8 MAC Design

8.1 Transmit and receive modes

The chip's transmitter has two separate operating modes, and the receiver has three separate operating modes.

Transmitter operating modes:

- Single packet mode
- Continuous mode

Receiver operating modes:

- Single packet mode
- Single packet mode with timeout
- Continuous mode

8.2 Transmitter Mode

In the MAC transmission mode, there are two modes: single packet transmission and continuous transmission. The available mode selection register can be used for selection. After the mode selection register selects the TX mode, the state machine enters the TX preparation state, waits for the FIFO to be filled with data, and then enters the TX transmission state.

When the MAC enters the TX transmission mode, the power consumption control module successively turns on the LDO module, PLL module, PA and other analog module circuits, and then sends the TX start signal to the Modem module to start sending data.

After the data is sent, the LDO module, PLL module, PA and other analog module circuits are turned off in turn. Then the IRQ signal is sent to the MCU. After the MCU clears the IRQ, the REG_OPT_MODE register can be configured to enter the STB3 mode and end a TX process.

8.2.1 Single packet mode

The state flow chart of the single packet sending mode is shown in the figure 8-1.

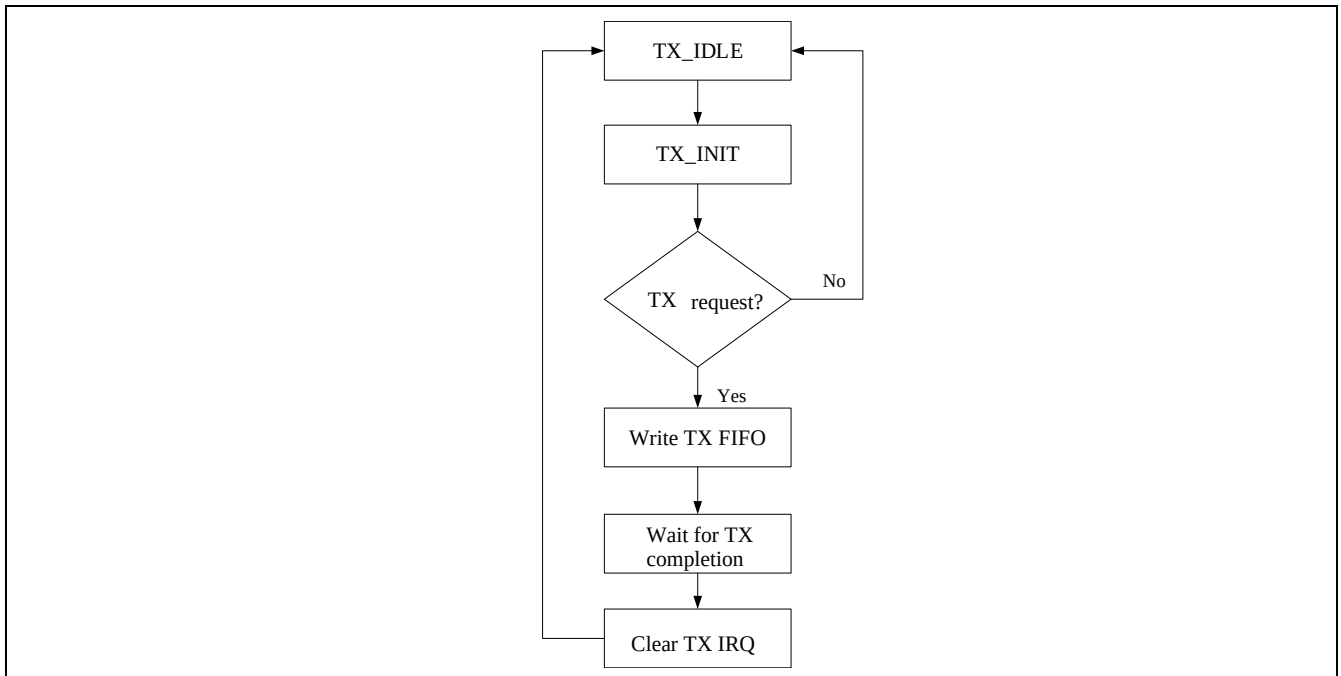


Figure 8-1 Single packet sending mode state flow chart

8.2.2 Continuous mode

The state flow chart of the continuous transmission mode is shown in Figure 8-2.

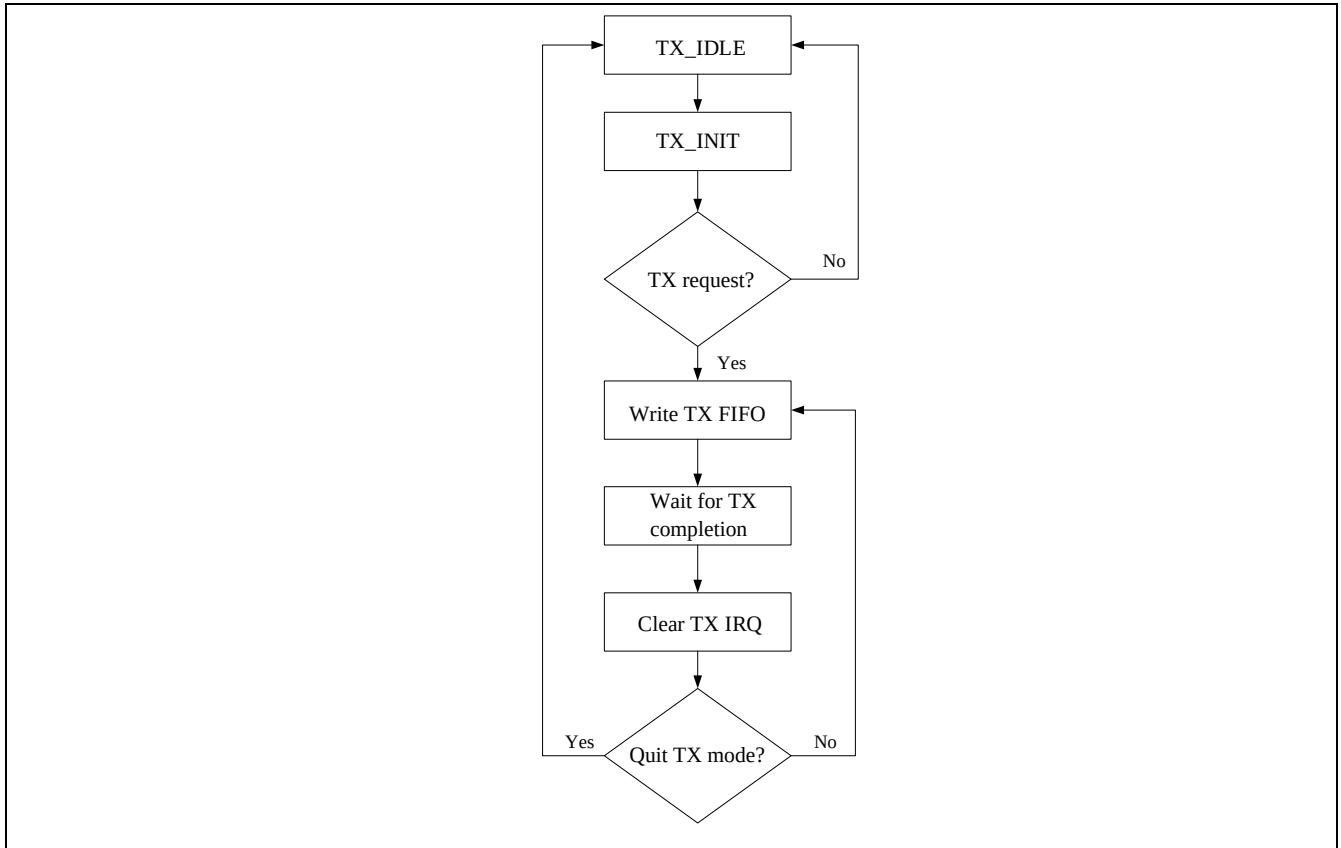


Figure 8-2 Continuous sending mode flow chart

8.3 Receiver Mode

In the MAC receiving mode, there are single packet receiving, single packet receiving with timeout, and continuous receiving modes, which are selected using a 2-bit mode selection register.

After the mode selection register selects the RX mode, the state machine enters the RX receiving state.

When the MAC enters the receiving mode, the power control module successively turns on analog module circuits such as LDO and PLL, and sends an RX start signal to the Modem to start receiving data.

After the data is received, analog module circuits such as LDO and PLL are turned off in turn. Then an IRQ signal is sent to the MCU. After the MCU clears the IRQ, the REG_OPT_MODE register can be configured to enter the STB3 mode and end an RX process.

8.3.1 Single packet mode

The state flow chart of the single packet receiving mode is shown in Figure 8-3.

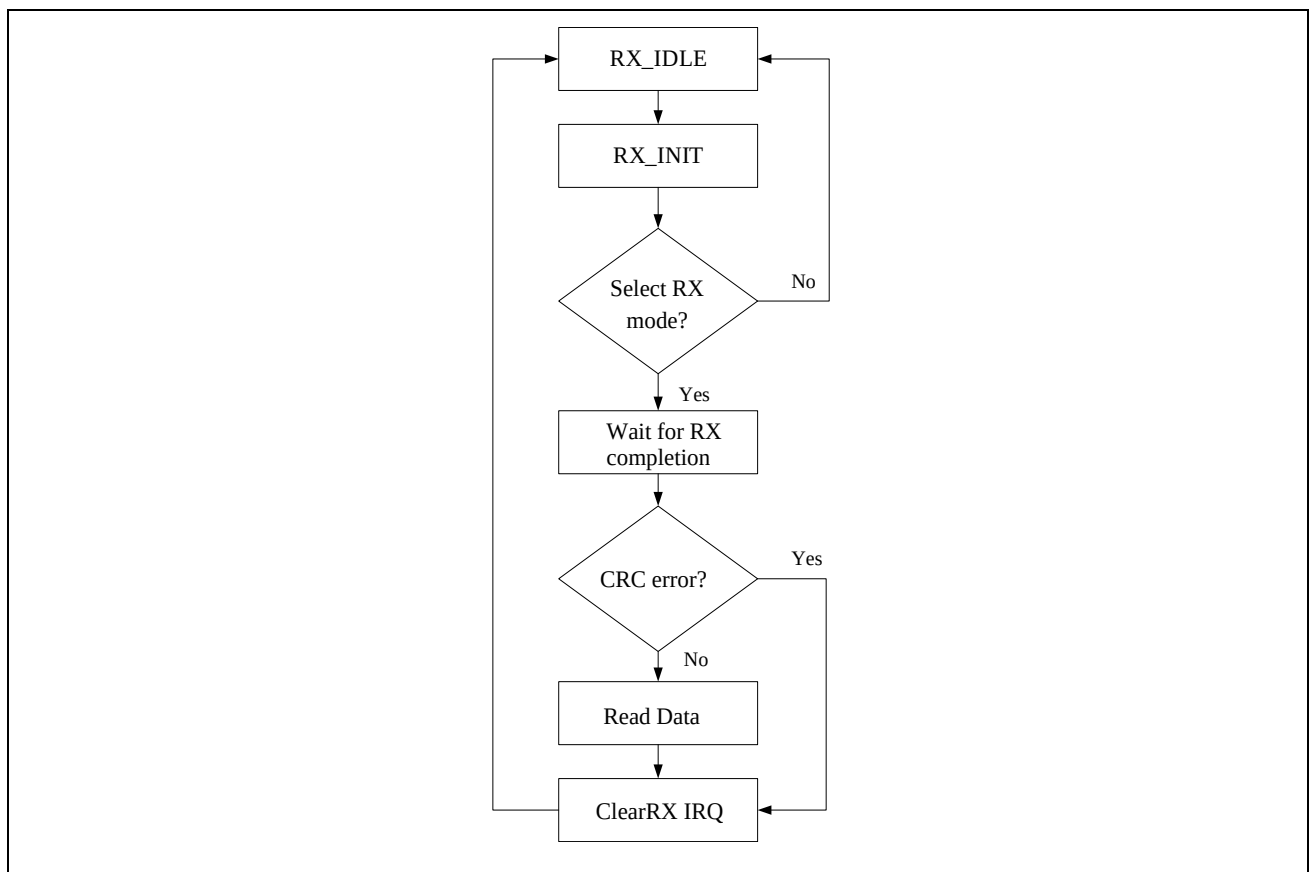


Figure 8-3 Flowchart of single packet receiving mode

8.3.2 Single packet mode with timeout

The state flow diagram of the single-packet receive mode with timeout is shown in Figure 8-4.

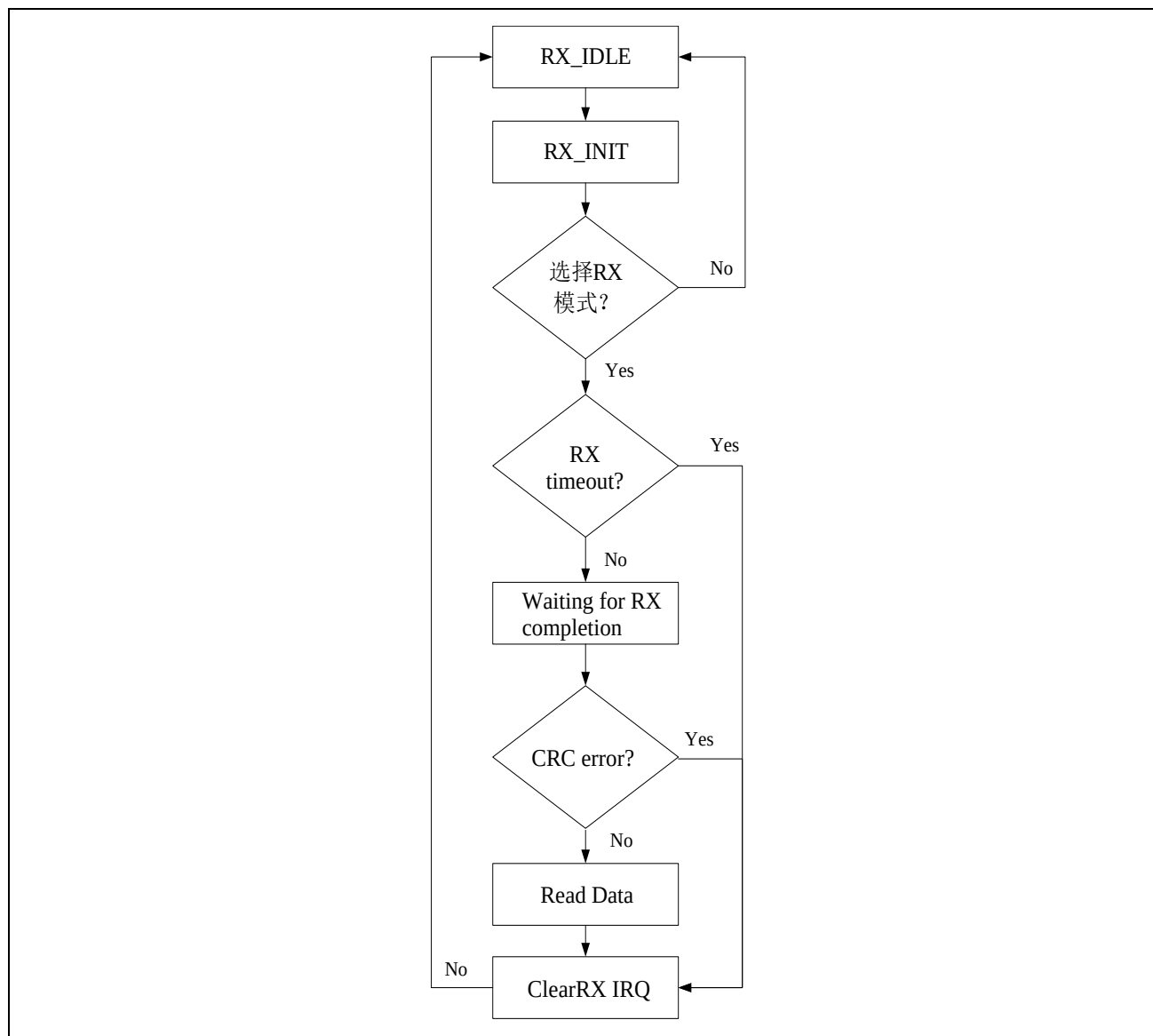


Figure 8-4 Flowchart of single packet receiving mode with timeout

8.3.3 Continuous mode

The state flow chart of the continuous receive mode is shown in Figure 8-5.

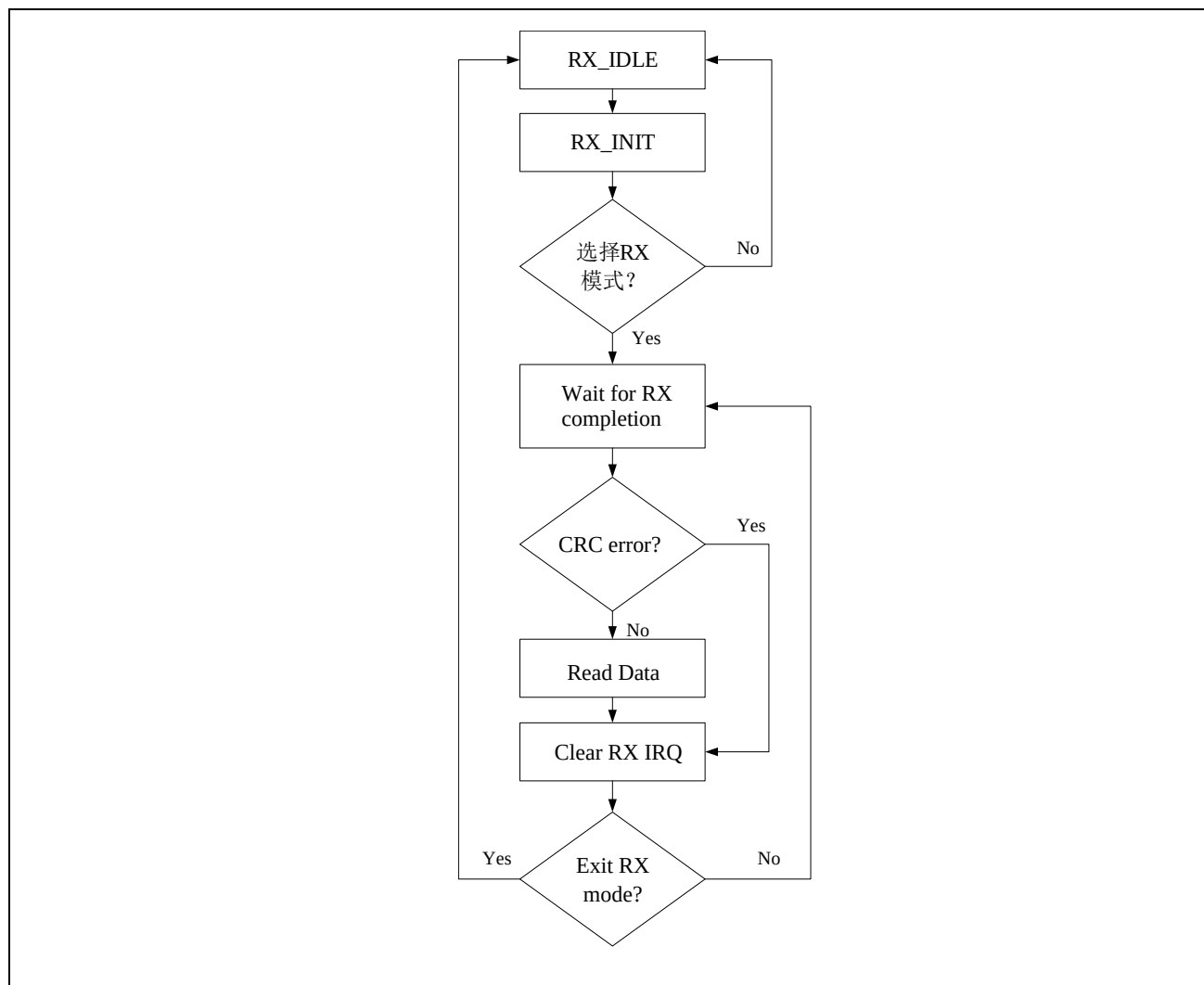


Figure 8-5 Continuous receiving mode flow chart

The difference between the continuous receiving mode and the single packet receiving mode and the single packet receiving mode with timeout is that after the data is received, an IRQ signal is sent to the MCU. After the MCU clears the IRQ, the next data reception begins. To exit the continuous receiving mode, the mode selection register selects the crystal oscillator output mode, and then turns off the LDO module, PLL module and other analog module circuits in turn, ending a continuous receiving mode process.



8.4 IRQ Interrupt

All interrupts of the chip share one external interrupt IO pin IRQ, and the interrupt pin is valid at high level. The chip has 7 types of interrupts, as follows. Interrupts can be enabled through the interrupt enable register, and the interrupt status can be read through the interrupt status register.

Interrupt Enable Register	Description
REG_MAPM_IRQ_MASK	MAPM 模式下的 IRQ MASK 使能 0: 产生 MAPM 的 IRQ 中断 1: 不产生 MAPM 的 IRQ 中断
REG_RX_HMP_ERR_MASK	纠错码错误 IRQ 中断掩码 0: 不产生纠错码错误的 IRQ 中断 1: 产生纠错码错误的 IRQ 中断
REG_RX_PLHD_DONE_MASK	提前中断模式掩码 0: 不产生提前中断模式的 IRQ 中断 1: 产生提前中断模式的 IRQ 中断
REG_RX_DONE_MASK	rx 完成 irq 掩码 0: 不产生 rx 完成的 IRQ 中断 1: 产生 rx 完成的 IRQ 中断
REG_RX_PL_CRC_ERR_MASK	Rx 有效负载 crc 错误 irq 掩码 0: 不产生 Rx 有效负载 crc 错误的 IRQ 中断 1: 产生 Rx 有效负载 crc 错误的 IRQ 中断
REG_RX_TIMEOUT_MASK	Rx 超时 irq 掩码 0: 不产生 Rx 超时的 IRQ 中断 1: 产生 Rx 超时的 IRQ 中断
REG_TX_DONE_MASK	Tx 完成 irq 掩码 0: 不产生 Tx 完成的 IRQ 中断 1: 产生 Tx 完成的 IRQ 中断

Interrupt Status Register	Description
REG_MDM_MAPM_IRQ	MAPM 的 IRQ 中断
REG_RX_HM_P_ERR_IRQ	纠错码错误的 IRQ 中断
REG_RX_PLHD_DONE_IRQ	提前中断模式的 IRQ 中断
REG_RX_DONE_IRQ	Rx 完成的 IRQ 中断
REG_RX_PL_CRC_ERR_IRQ	Rx 有效负载 crc 错误的 IRQ 中断
REG_RX_TIMEOUT_IRQ	Rx 超时的 IRQ 中断
REG_TX_DONE_IRQ	Tx 完成的 IRQ 中断



8.5 CAD-IRQ Interrupt

The chip CAD-IRQ interrupt pin reuses the GPIO11 pin. When using it, you need to configure the GPIO11 hardware to CAD-IRQ output mode.

Register Name	Register Location	Description
REG_OUT_FUNC_SEL_CAD	page0, 0x5e, bit6	CAD 功能选择 当 REG_OUT_FUNC_SEL_CAD 为 1, GPIO 为普通的 IO, 其输出可由寄存器 PAD_OUT_REG 配置; 当 REG_OUT_FUNC_SEL_CAD 为 0, GPIO 输出为 CAD 功能管脚。
PAD_OUT_REG_CAD	page0, 0x68, bit3	控制 GPIO11 输出的寄存器。



9 Interface design

The chip can read and write registers and transceiver FIFOs through four-wire SPI, three-wire SPI and I2C. By default, it supports both four-wire SPI and I2C. The three-wire SPI needs to be configured through the register SPI3_EN_REG to take effect.

Note: FIFO address is a special area. The register address of this area is the entry address of these special areas. The operation method is different from that of regular registers. For the FIFO operation method, please refer to Chapter 9.4 FIFO.

9.1 Four-wire SPI

The chip implements the SPI bus slave, which is used to read and write registers and FIFO. The SPI bus is a four-wire system, which are:

- SCK (clock)
- CSN (Chip select signal, low level active)
- MOSI (Data input)
- MISO (Data output)

Among them, SCK, CSN, and MOSI are controlled by the host Master, and MISO is controlled by the Slave.

During the communication process, the CSN level is pulled low to start, and the transmission process ends when the CSN level is pulled high. The host Master sends data through MOSI and MISO receives data. Data is generated at the falling edge of SCK and data sampling is performed at the rising edge.

The information transmitted by the Master consists of two parts: Address Byte and Data Byte. The first 7 bits of Address Byte are the address bits `addr`; the last 1 bit is the read/write bit `wr`, which is set to 1 during a write operation and set to 0 during a read operation.

SPI has three transmission modes:

- Single: Single byte transmission mode. The information is only 2 bytes, and the Master sends the Address Byte through MOSI.

If it is a write operation, the Master continues to send the Data Byte through MOSI; if it is a read operation, the Master reads the Data Byte replied by the Slave on MISO.

- Burst: Burst continuous transmission mode. When the information is larger than 2 bytes, the Address Byte is followed by several Data Bytes. There is no need to add Address Bytes between Data Bytes. The Slave will automatically increment the address between each Data Byte. The CSN signal is pulled high after the last Data Byte, and the rest of the information transmission process is maintained at a low level.
- FIFO: FIFO read/write mode. In this mode, single byte or continuous transmission can be realized. The transmission rules are the same as Single mode and Burst mode. The difference is that the address bit `addr` in Address Byte can only be configured as 0x01, and Slave does not perform address increment operation between Data Bytes.

In addition, by configuring the REG_OUT_PAD_MODE register, MISO can be configured to input high-impedance state to achieve a 4-wire SPI one-master-multiple-slave connection architecture.



9.1.1 Four-wire SPI write timing

The SPI write timing is as follows:

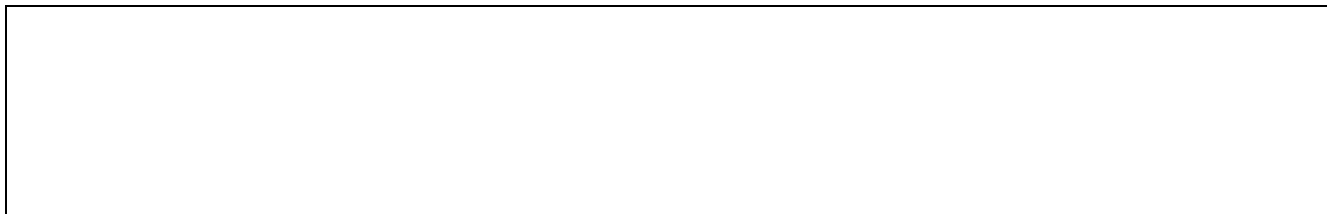


Figure 9-1 SPI write timing

9.1.2 Four-wire SPI read timing

The SPI read timing is as follows:

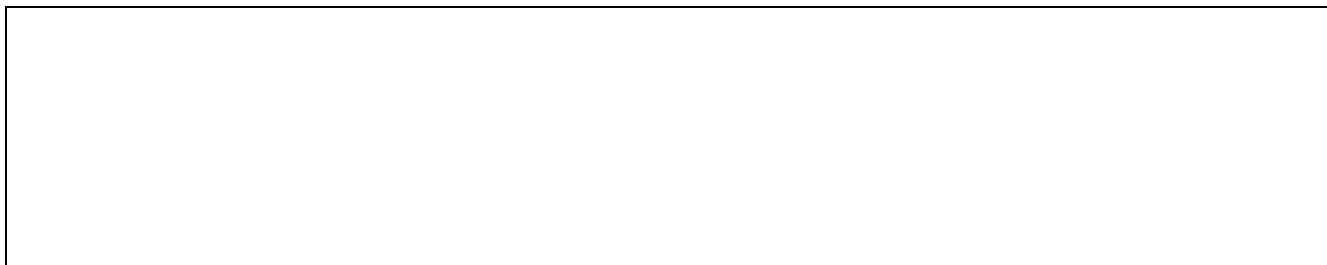


Figure 9-2 SPI read timing

9.1.3 Four-wire SPI timing requirements

All SPI timing requirements in the table below are as follows:

Parameter	Description	Minimum	Typical	Maximum	Unit
t1	SCN 下降沿到 SCK 上升沿	32	-	-	ns
t2	SCK 时钟周期	100	-	-	ns
t3	SCK 高电平时间	50	-	-	ns
t4	MOSI 到 SCK 时钟的 SETUP 时间	7	-	-	ns
t5	MOSI 到 SCK 时钟的 HOLD 时间	7	-	-	ns
t6	SCK 下降沿到 MISO 延时	0	-	15	ns
t7	SCK 到 SCN 上升沿时间	0	-	150	ns



9.2 Three-wire SPI

The chip supports four-wire SPI and I2C by default. If you need to use three-wire SPI, you need to configure the register SPI3_EN_REG after power-on to make the three-wire SPI effective. The three-wire SPI signals include:

- SCK (clock)
- CSN (chip select signal, low level active)
- MOSI (Data Input/Output)

9.2.1 Three-wire SPI write timing

The SPI write timing is as follows:

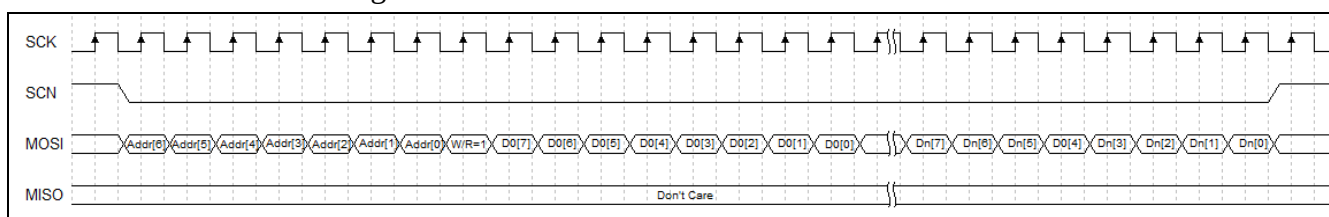


Figure 9-3 3-wire SPI write timing

9.2.2 Three-wire SPI read timing

The SPI read timing is as follows:

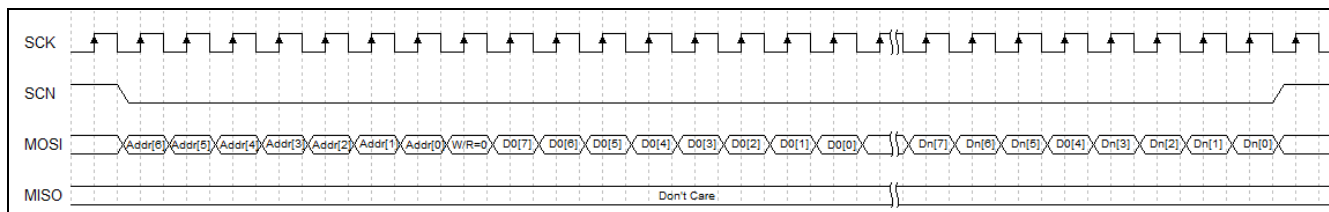


Figure 9-4 3-wire SPI read timing

9.2.3 Three-wire SPI timing requirements

Please refer to Section 9.1.3.



9.3 I2C

The chip supports I2C by default. I2C signals SCL and SDA multiplex SPI's SCK and MOSI signals.

Note: When using I2C, the CSN signal is required to be high.

The I2C device address width is 7 bits and the value is 0x72.

The timing diagram of the I2C start signal Start is as follows:

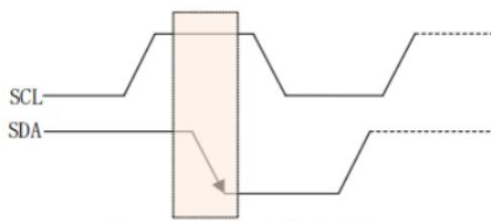


Figure 9-5 I2C start signal

The timing diagram of the I2C termination signal Stop is as follows:

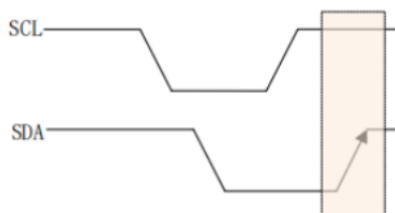


Figure 9-6 I2C stop signal

9.3.1 I2C Write Timing

The I2C write timing is as follows:

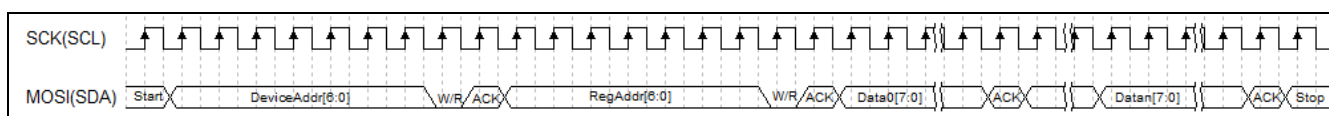


Figure 9-7 I2C write timing

9.3.2 I2C Read Timing

The I2C read timing is as follows:

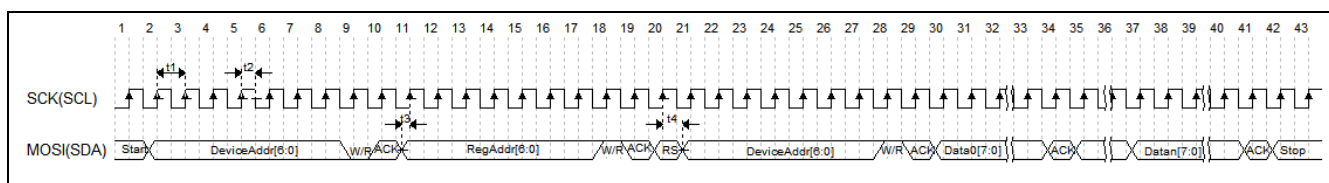


Figure 9-8 I2C read timing



9.3.3 I2C Timing Requirements

Parameter	Description	Minimum	Typical	Maximum	Unit
t1	SCL 时钟周期	200	-	-	ns
t2	SCL 高电平时间	100	-	-	ns
t3	SDA 到 SCL 时钟的 SETUP 时间	12	-	-	ns
t4	SDA 到 SCL 时钟的 HOLD 时间	12	-	-	ns

9.4 FIFO

The chip has a 256-byte FIFO to store the TX module's transmitted data and the RX module's de-coded data.

The FIFO consists of a single-port RAM and can only store and read a single packet of data information. If there is already a packet of data in the FIFO, the data should be read before writing, otherwise the previous packet of data in the FIFO will be overwritten.

In the working mode of STB3 and later, the FIFO can be read and written by Modem, SPI, and I2C.

The read and write address of the FIFO is 0x01, and the FIFO content can be read and written through this address through SPI or I2C.

9.5 GPIO

The chip supports multiple GPIO input and output controls, and the GPIO can be controlled through register configuration. For register configuration, please refer to Section 12.4. Each GPIO port can be configured as internal pull-up/pull-down input, high-impedance input (floating input), push-pull output (CMOS output) and other modes. The configuration method is shown in Table 9-1.

Table 9-1 GPIO port mode configuration

DIEN	OE	OUT	PUEN	PDEN	IO direction	IO state
0	1	0/1	0	0	推挽输出	0/1
0	0	x	0	0	开漏输出	高阻
1	0	x	0	0	输入	高阻
1	0	x	1	0	输入	上拉
1	0	x	0	1	输入	下拉

Note: The output modes of GPIO10 and GPIO11 in the chip can be reused to implement special functions.

9.5.1 External PA control GPIO

GPIO10 has two configuration output modes: normal IO and PA control pin. The different modes of GPIO10 can be configured through the register REG_OUT_FUNC_SEL_PA.

When GPIO10 is configured as a normal IO, it can output high and low levels, just like other GPIOs.

When GPIO10 is configured as PA control, and the external control PA mode is configured to the on state through the register, GPIO10 can be used as the enable control pin of the external PA, that is, GPIO10 only outputs a high level when sending data in the transmission mode, and outputs a low level at other times.



Register Name	Register Location	Description
REG_OUT_FUNC_SEL_PA	page0, 0x5e, bit7	GPIO10 功能选择 当 REG_OUT_FUNC_SEL_PA 为 1, GPIO10 为普通的 IO, 其输出可由寄存器 PAD_OUT_REG_PA 配置; 当 REG_OUT_FUNC_SEL_PA 为 0, GPIO10 输出为外置 PA 功能管脚。
PAD_OUT_REG_PA	page0, 0x68, bit2	控制 GPIO10 输出的寄存器。 当 REG_OUT_FUNC_SEL_PA 为 1, GPIO10 为普通的 IO, 其输出可由寄存器 PAD_OUT_REG_PA 配置。

9.5.2 CAD monitoring GPIO

GPIO11 has two configuration modes: normal IO and CAD detection pin. The different modes of GPIO11 can be configured through the register REG_OUT_FUNC_SEL_CAD. When GPIO11 is configured as a normal IO, it can output high and low levels, just like other GPIOs.

When GPIO11 is configured as a CAD signal detection interrupt output, and the CAD function is also configured to be turned on through the register, and the chip is in the RX state, when receiving a signal with the same bandwidth and spreading factor parameters as the currently configured, GPIO11 will generate a continuous and stable high level, indicating that the chip has searched for an air signal, and the high level will not be pulled down until the signal disappears; in other situations and modes, GPIO11 outputs a low level. For specific CAD configuration and use, please refer to Chapter 9.4.

The chip CAD-IRQ interrupt pin reuses the GPIO11 pin. When using it, you need to configure the GPIO11 hardware to CAD-IRQ output mode.

Register Name	Register Location	Description
REG_OUT_FUNC_SEL_CAD	page0, 0x5e, bit6	CAD 功能选择 当 REG_OUT_FUNC_SEL_CAD 为 1, GPIO 为普通的 IO, 其输出可由寄存器 PAD_OUT_REG 配置; 当 REG_OUT_FUNC_SEL_CAD 为 0, GPIO 输出为 CAD 功能管脚。
PAD_OUT_REG_CAD	page0, 0x68, bit3	控制 GPIO11 输出的寄存器。 当 REG_OUT_FUNC_SEL_CAD 为 1, GPIO11 为普通的 IO, 其输出可由寄存器 PAD_OUT_REG_CAD 配置。



10 ChirpIoT special function description

10.1 RSSI Function

10.1.1 ChirpIoT signal RSSI indication

The chip supports the signal energy statistics function. When the chip receives ChirpIoT data, the internal logic of the chip will count the signal strength RSSI value of the current data packet. However, after the data packet is received, the MCU can read the register RSSI_R32_HOLD or RSSI_RBW_HOLD through the SPI interface to obtain the RSSI value of the current data packet.

10.1.2 Channel detection RSSI indication

The RSSI function also supports energy statistics indication when detecting channels. After the chip enters the RX state, it can perform RSSI energy statistics once by configuring the register REG_RSSI_PRE_RD, and then read the current signal RSSI energy through the register RSSI_R32_HOLD.

REG_RSSI_PRE_RD is first configured to 0 and then to 1 to update the RSSI value once. The register location is as follows:

Register Name	Register Location	Description
REG_RSSI_PRE_RD	page2, 0x06, bit2	检测信道 RSSI 读取开关，由 0 配置为 1 时更新一次 RSSI 值。

10.2 Signal-to-noise energy statistics function

The chip supports signal noise statistics. When the chip receives ChirpIoT data, the internal logic of the chip will calculate the signal energy and noise energy of the current data packet. However, after the data packet is received, the MCU can read the signal energy value of the register SIG_POW_AVG and the noise energy value of the register NOI_POW_AVG through the SPI interface, and then the MCU can calculate the signal-to-noise ratio of the data packet through the standard formula.

10.3 Channel Activity Detection (CAD)

Since ChirpIoT™ signals have excellent receiving performance, they can still ensure normal signal reception and demodulation when the signal strength is lower than the receiver noise floor. In this case, the method of using only the signal strength indicator RSSI to determine whether the channel is occupied can no longer meet the needs. Therefore, the chip is specially designed with a channel activity detector to detect ChirpIoT™ signals, ensuring that the channel can be found to be occupied by ChirpIoT™ signals sent by other devices when the signal strength is lower than the receiver noise floor. The detection function of the channel interaction detector is divided into two parts, one is the ChirpIoT™ signal preamble detection, and the other is the ChirpIoT™ signal data segment detection.



10.3.1 ChirpIoT signal leading segment detection

As shown in Figure 10-1:

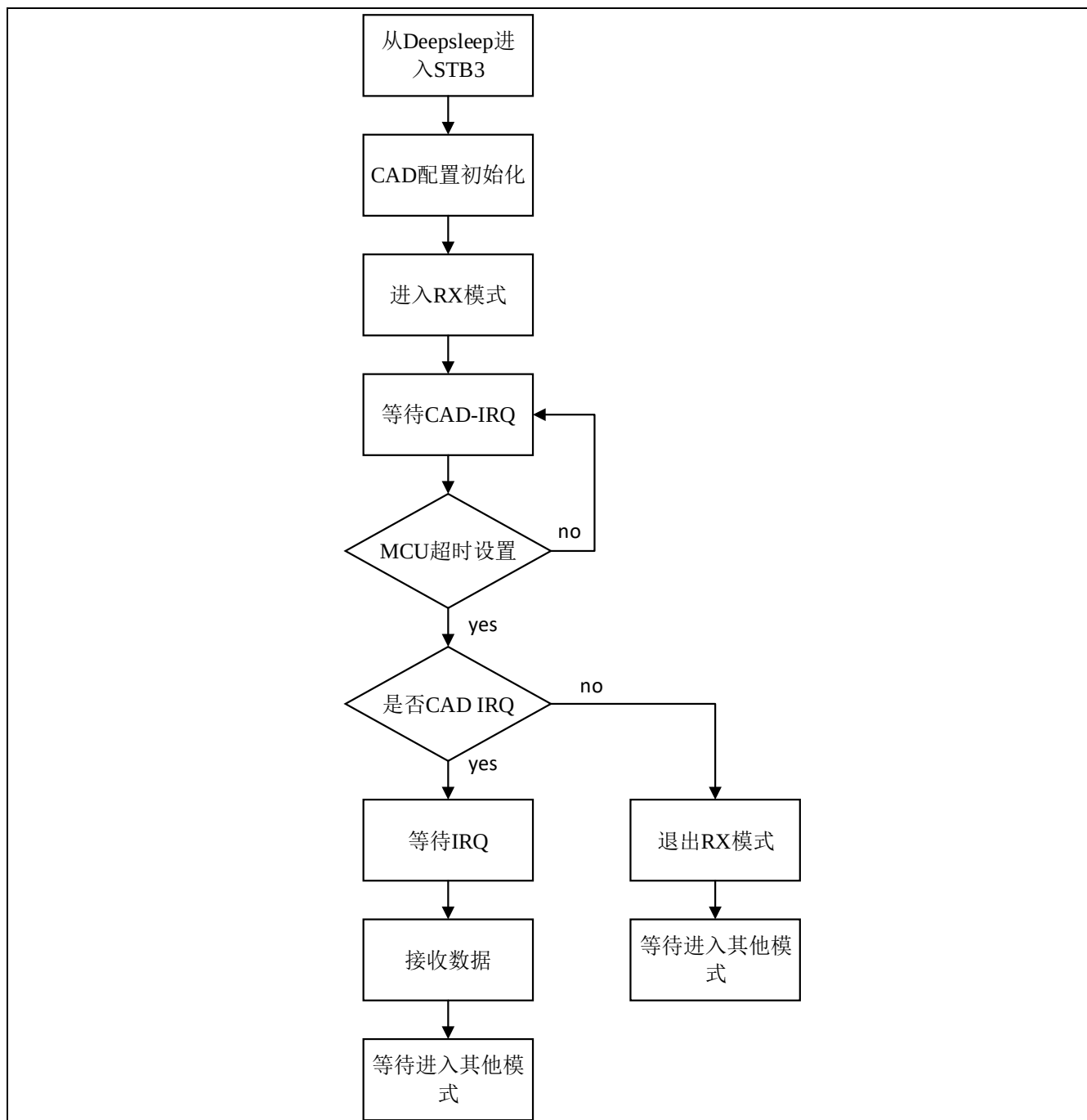


Figure 10-1 Channel Activity Detection (CAD)

The chip supports CAD-IRQ interrupt. After entering RX mode, the chip will detect whether there is a leading signal of ChirpIoT™ in the channel. If so, CAD-IRQ will be set high. The external MCU can determine whether there is a leading signal of ChirpIoT™ in the channel by detecting whether the CAD-IRQ signal is pulled high within a certain period of time. If so, it can continue to wait for the IRQ interrupt of data reception and then receive data; if there is no leading signal of ChirpIoT™,



则快速退出 RX 模式，等待进入其他模式命令。

另外，为了减少 CAD-IRQ 的误触发，增加了 CAD-IRQ 的滤波功能，通过配置寄存器可启动该功能。

Address	Register	Bit	Default	Access	Description
[1][0x25]	REG_DEC_VAL_THRE	1:0	0x3	W/R	CAD 检测阈值，规定连续接收到几个 preamble 之后 CAD 拉高。

10.3.2 ChirpIoT signal data segment detection

为了避免错过 ChirpIoT 信号的前导信号而导致实际接收过程中无法实现有效 CAD 检测，芯片还同时支持非 ChirpIoT 前导信号的检测，即 ChirpIoT 数据段信号检测功能。通过配置寄存器 REG_SIGNAL_POWER_THRED、REG_DEC_PLD_THRE、REG_DEC_PLD_EN 可实现启动数据段信号检测。

寄存器配置说明：

Address	Register	Bit	Default	Access	Description
[1][0x35]	REG_SIGNAL_POWER_THRED	7:4	4'b1110	W/R	CAD 有效信号的阈值，只有能量大于此值才能进入 CAD 有效信号的判断。
	REG_DEC_PLD_THRE	3:2	2'b10	W/R	CAD 有效信号个数阈值，只有接收到满足条件的此数值个数的 preamble，才能拉高 CAD。
	REG_DEC_PLD_EN	1	1'b0	W/R	CAD payload 检测使能 0: disable 1: enable

注意：前导段和数据段的 CAD 检测可以同时打开，确保不同阶段数据可检测判断。

10.4 MAPM function

在 ChirpIoT 星状组网使用场景中，为了保证节点端设备能收到网关发送的数据，网关端会使用超长 Preamble 进行发送，如 preamble 占用时间 1 秒。在这种情况下，所有节点端设备在收到 preamble 后都会一直接收直到 payload 阶段显示节点网络地址，此时，节点端设备的软件可以比较收到的网络地址和节点设备自己地址，如不同，则停止接收，降低功耗，如满足地址接收条件则继续接收。这种方案存在几个问题：

- ① 不满足地址接收条件的节点端设备，直到 payload 阶段才能停止接收，浪费大量功耗
- ② 满足地址接收条件的节点端设备也需要收满超长的 preamble 才能进入 payload 接收阶段，也较浪费功耗

为解决以上两个功耗问题，芯片增加新的帧结构模式 Multiple Address Preamble Mode (MAPM)。该帧结构模式是在 Preamble 阶段插入特殊地址，这样节点端在 Preamble 阶段就可以根据地址来提前判断是否满足地址接收条件，已解决①问题。另外，为了



解决第②问题，在特殊地址中增加可变计数器，表明后续还有多少 Preamble 长度，满足地址接收条件的节点端设备可提前进入低功耗休眠状态，后根据该计数器判断需要提前多久在进入接收状态，保证数据正常接收，以便进一步节省满足地址接收条件的节点端设备功耗。

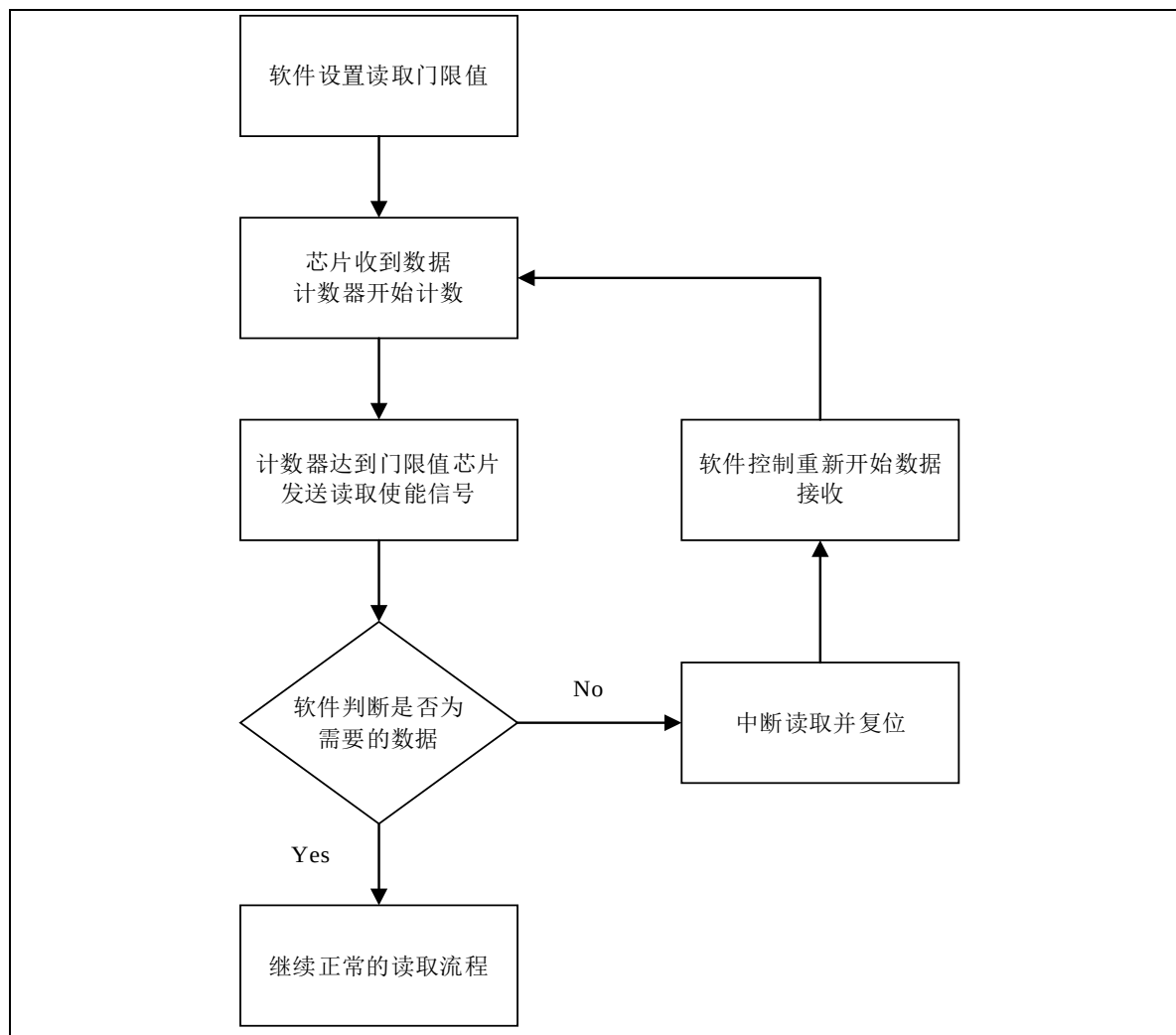
10.5 eFuse Function

芯片支持 82Bytes 的 eFuse 空间区域供客户使用，可用于存储掉电不丢失的数据。

10.6 Early interrupt function

芯片提前中断功能是在芯片读取一帧数据的过程中，查看已经解出来的数据，判断是不是自己想要的，再决定继续读取还是放弃这帧数据。

流程图如下所示：





10.7 Intelligent SF recognition

芯片为实现轻量化网关设备，设计了智能化 SF 识别硬件电路，可完成在同一个频点根据实际信道中的 ChirpIoT™的 SF 模式，智能化识别，并自动配置芯片 SF 参数，达到接收不同 SF 信号数据的目的。

具体实现流程如图 10-2 所示：

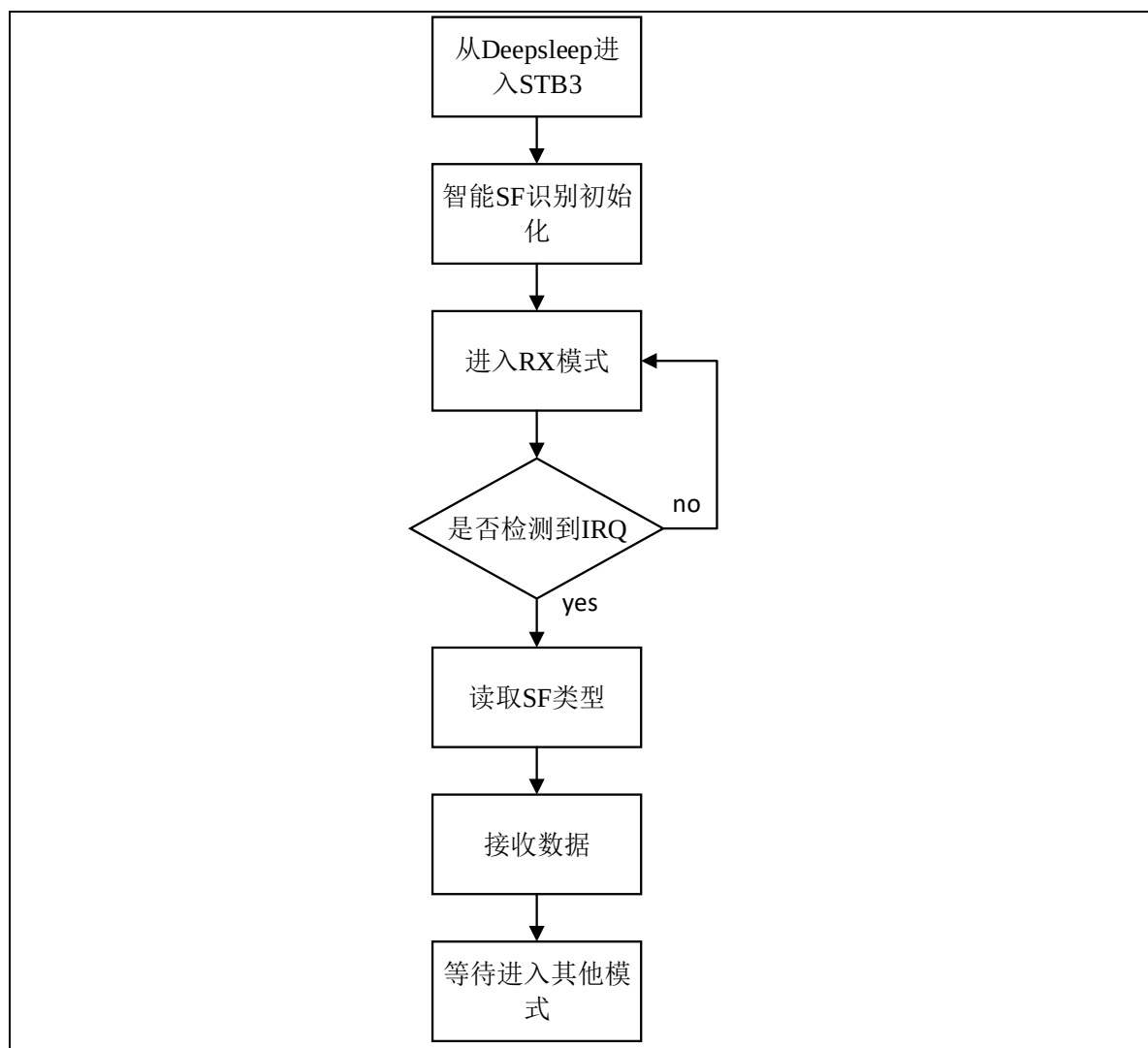


图 10-2 智能化 SF 识别



11 Operation Mode Description

11.1 Overview

芯片一共有 7 种模式：DeepSleep 模式、Sleep 模式、STB1 模式、STB2 模式、STB3 模式、TX 模式和 RX 模式。

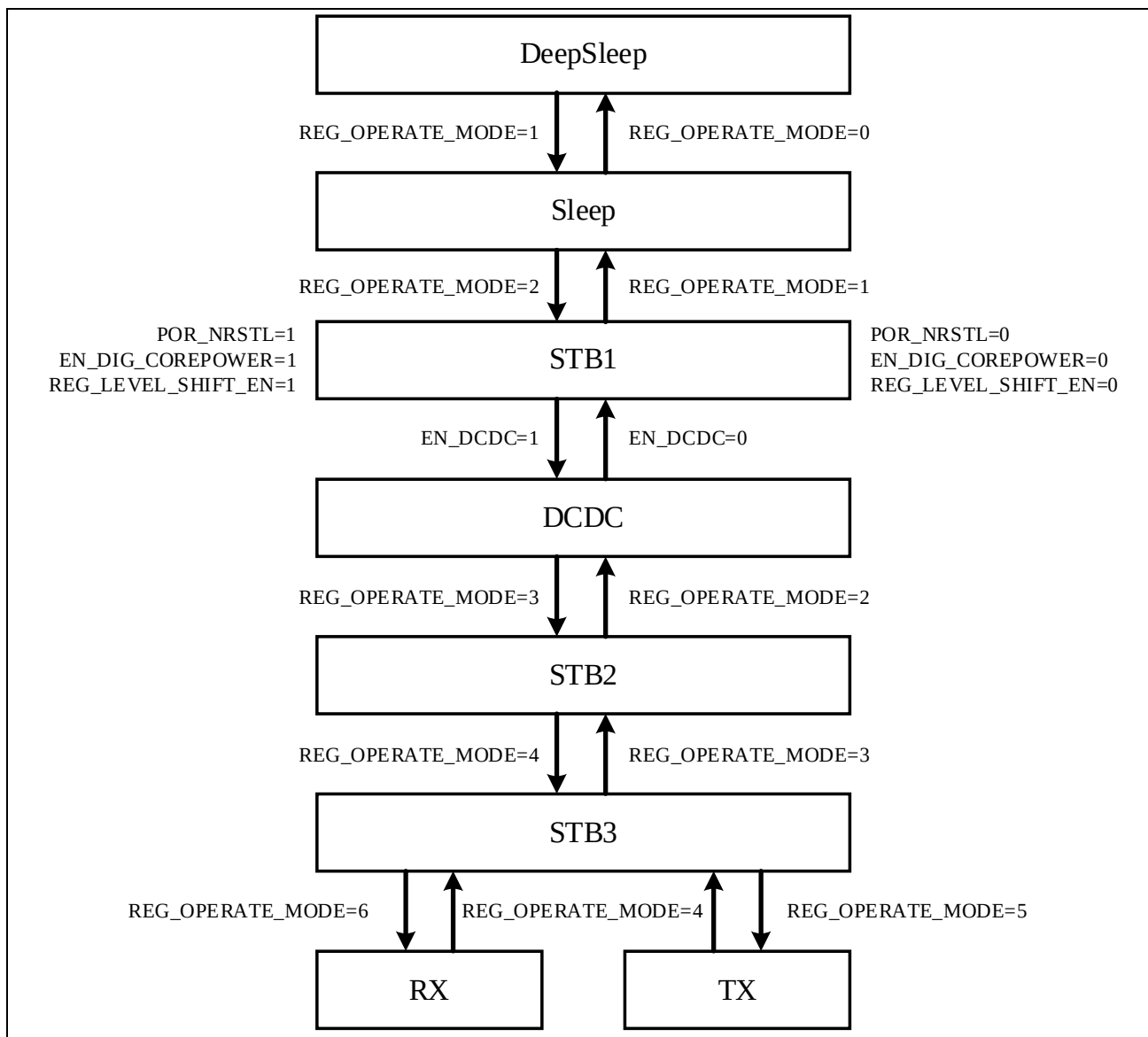


图 11-1 工作模式设置流程

芯片上电后有七种状态，包括深度睡眠模式(DeepSleep)、睡眠模式(Sleep)、LDO 工作模式(STB1)、OSC 工作模式(STB2)、OSC 输出模式(STB3)、TX 模式以及 RX 模式。所有状态可通过配置寄存器进行切换。

注：如果芯片外围电路不支持 DCDC，可不配置 EN_DCDC，直接从 STB1 进入 STB2。芯片电路可参考第 13 章。



11.2 Working modes description

11.2.1 Deep Sleep mode

该模式下 3V 逻辑区域工作，除 3V 逻辑区域寄存器可配置，其余模块不可操作，保持芯片功耗最小。

11.2.2 Sleep mode

该模式下使能低功耗 LDO，芯片功耗较低，且可支持所有寄存器保电，但禁止通过 SPI/I2C 配置除 3V 逻辑区域以外的其他寄存器。

11.2.3 STB1 mode

该模式高性能 LDO 开始工作，低压区域逻辑使能。

11.2.4 DCDC mode

在 STB1 状态下可通过给 EN_DCDC 寄存器写入高电平来打开芯片的 DCDC 功能。

注意：DCDC 模式需要相关外围电路支持，请参考第 13 章。

11.2.5 STB2 mode

该模式打开 OSC 晶振，OSC 开始工作维持震荡但不输出给其他模块，功耗相对较小。

11.2.6 STB3 mode

该模式下 OSC 时钟输出给各模块，芯片开始正常工作。

11.2.7 TX Mode

该模式已打开所有 TX 发射相关的模块，可实现包数据的编码和发送。

11.2.8 RX Mode

该模式已打开所有 RX 接收相关的模块，等待包数据的接收和解码。



12 Registers

The chip registers are divided into 4 pages, each page has 128 register addresses. The register page is selected through the REG_PAGE_SEL of register 0x00. The registers at addresses 0x00~0x05 can be operated in any register page. The following table describes the location and function of each register of the chip.

Note:

Reading and writing of the registers in the Reserved section are prohibited.

All registers in the low-voltage area only support operation in STB1, STB2, STB3, Tx and Rx modes.

12.1 System Configuration

Page	Addr.	Register	Bit	Default	Access	Description
0/1/2/3	0x00	REG_SOFT_RST	7	0x00	W/R	Soft reset, except 1.2 V configuration registers 1.2 V Logic reset 0: Reset 1: No reset
		Reserved	6:2		W/R	Reserved
		REG_PAGE_SEL	1:0		W/R	0: Page0 1: Page1 2: Page2 3: Page3
	0x01	FIFO READ/ WRITE ACCESS POINT	7:0	-	W/R	FIFO Read and write address

12.2 Mode Configuration

Page	Addr.	Register	Bit	Default	Access	Description
0/1/2/3	0x02	Reserved	7:3	0x00	W/R	Reserved
		REG_OPERATE_MODE	2:0		W/R	0: 深度睡眠模式 1: 睡眠模式 2: STB1 模式(LDO 工作) 3: STB2 模式(OSC 工作) 4: STB3 模式(OSC 输出) 5: Tx 模式 6: Rx 模式
	0x04	Reserved	7:6	0x06	W/R	Reserved
		EN_LS_3V	5		W/R	1: 电平转换使能
		POR_RSTL	4		W/R	1.2V 所有配置寄存器复位 0: 复位 1: 不复位
		Reserved	3:0		W/R	Reserved



12.3 MAC configuration (low voltage area registers)

Page	Addr.	Register	Bit	Default	Access	Description
0	0x54	Reserved	7:3	0x00	W/R	Reserved
		REG_TX_FIFO_FULL	2		W/R	0: TX FIFO 未 1: TX FIFO 已
		REG_TX_FIFO_FULL_SEL	1		W/R	0: select fifo wr done 1: select reg tx fifo full
		Reserved	0		W/R	Reserved
	0x58	REG_RX_TIMEOUT_IMMED	7	0x0F	W/R	1: Rx 超时 IRQ 立即生成
		REG_MAPM_IRQ_MASK	6		W/R	MAPM 模式下的 IRQ MASK 使能 0: 不产生 MAPM 的 IRQ 中断 1: 产生 MAPM 的 IRQ 中断
		Reserved	5		W/R	Reserved
		REG_RX_PLHD_DONE_MASK	4		W/R	提前中断模式掩码 0: 不产生提前中断模式的 IRQ 中断 1: 产生提前中断模式的 IRQ 中断
		REG_RX_DONE_MASK	3		W/R	rx 完成 irq 掩码 0: 不产生 rx 完成的 IRQ 中断 1: 产生 rx 完成的 IRQ 中断
		REG_RX_PL_CRC_ERR_MASK	2		W/R	Rx 有效负载 crc 错误 irq 掩码 0: 不产生 Rx 有效负载 crc 错误的 IRQ 中断 1: 产生 Rx 有效负载 crc 错误的 IRQ 中断
		REG_RX_TIMEOUT_MASK	1		W/R	Rx 超时 irq 掩码 0: 不产生 Rx 超时的 IRQ 中断 1: 产生 Rx 超时的 IRQ 中断
		REG_TX_DONE_MASK	0		W/R	Tx 完成 irq 掩码 0: 不产生 Tx 完成的 IRQ 中断 1: 产生 Tx 完成的 IRQ 中断
	0x5b	Reserved	7:3	0x03	W/R	Reserved
		REG_RX_DONE_RST_TEST	2		W/R	1: 一旦 RX 完成, modem 即刻复位 0: IRQ 清零且 RX 完成后 modem 才复位
		REG_RX_CR_ERR_RST	1		W/R	1: rx 码率错误复位使能
		REG_RX_HEADER_ER_RST	0		W/R	1: rx 包头错误复位使能
	0x5d	Reserved	7:1	0x00	W/R	Reserved
		REG_2POINT_CAL_EN	0		W/R	两点 cal 使能, 高电平有效
	0x6c	EFUSE_WR_IRQ	7	0x00	R	eFuse 读写完中断 1: 高电平说明读写完成
		REG_MDM_MAPM_IRQ	6		R	MAPM 中断状态寄存器 1: MAPM IRQ; 0: 非 MAPM IRQ
		Reserved	5		W/R	Reserved
		REG_RX_PLHD_DONE_IRQ	4		W/R	写 1 清零提前中断
		REG_RX_DONE_IRQ	3		W/R	写 1 清零 rx done irq
		REG_RX_PL_CRC_ERR_IRQ	2		W/R	写 1 清零 rx payload crc error irq



		REG_RX_TIMEOUT_IRQ	1		W/R	写 1 清零 rx timeout irq
		REG_TX_DONE_IRQ	0		W/R	写 1 清零 tx done irq
	0x6D	REG_MAPM_PN_CNT	7:0	0x00	R	MAPM 模式的 preamble 数量计数器
	0x6E	REG_MAPM_ADDRX	7:0	0x00	R	MAPM 模式的 addrx

12.4 GPIO interface configuration (low voltage area registers)

Page	Addr.	Register	Bit	Default	Access	Description
0	0x5e	REG_OUT_FUNC_SEL_PA	7	0x00	W/R	1: 作为普通 IO 使用 0: 作为外置 PA 使用
		REG_OUT_FUNC_SEL_CAD	6		W/R	1: 作为普通 IO 使用 0: 作为 CAD 引脚使用
		Reserved	5:0		W/R	Reserved
	0x5f	GPIO_PDEN LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 下拉使能 GPIO_PDEN LSB[0]: GPIO0 下拉使能 其他配置 Reserved
	0x60	Reserved	7:4	0x00	W/R	Reserved
		GPIO_PDEN MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 下拉使能 GPIO_PDEN MSB[2]: GPIO10 下拉使能 其他配置 Reserved
	0x61	GPIO_PUEN LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 上拉使能 GPIO_PDEN LSB[0]: GPIO0 上拉使能 其他配置 Reserved
	0x62	Reserved	7:4	0x00	W/R	Reserved
		GPIO_PUEN MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 上拉使能 GPIO_PDEN MSB[2]: GPIO10 上拉使能 其他配置 Reserved
	0x63	GPIO_DIEN LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 输入使能 GPIO_PDEN LSB[0]: GPIO0 输入使能 其他配置 Reserved
	0x64	Reserved	7:4	0x00	W/R	Reserved
		GPIO_DIEN MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 输入使能 GPIO_PDEN MSB[2]: GPIO10 输入使能 其他配置 Reserved
	0x65	GPIO_OE LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 输出使能 GPIO_PDEN LSB[0]: GPIO0 输出使能 其他配置 Reserved
	0x66	Reserved	7:4	0x00	W/R	Reserved
		GPIO_OE MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 输出使能 GPIO_PDEN MSB[2]: GPIO10 输出使能 其他配置 Reserved
	0x67	GPIO_OUT LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 输出 GPIO_PDEN LSB[0]: GPIO0 输出 其他配置 Reserved
	0x68	Reserved	7:4	0x00	W/R	Reserved
		GPIO_OUT MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 输出



						GPIO_PDEN MSB[2]: GPIO10 输出 其他配置 Reserved
	0x74	Reserved	7:6	-	R	Reserved
		GPIO_IN LSB	5:0	-	R	GPIO_PDEN LSB[3]: GPIO3 输入 GPIO_PDEN LSB[0]: GPIO0 输入 其他配置 Reserved
	0x75	Reserved	7:6	-	R	Reserved
		GPIO_IN MSB	5:0	-	R	GPIO_PDEN LSB[3]: GPIO11 输入 GPIO_PDEN LSB[2]: GPIO10 输入 其他配置 Reserved

12.5 Basic operation configuration

Page	Addr.	Register	Bit	Default	Access	Description
3	0x06	Reserved	7:3	0x00	W/R	Reserved
		REG_TX_CFG_MODE	2		W/R	0: 单次发送模式 1: 连续发送模式
		REG_RX_CFG_MODE	1:0		W/R	0: 单次接收模式 1: 带超时的单次接收模式 2: 连续接收模式
	0x07	REG_RX_TIMEOUT[7:0]	7:0	0x00	W/R	接收超时设置, 低 8 位, 单位微秒
	0x08	REG_RX_TIMEOUT[15:8]	7:0	0x00	W/R	接收超时设置, 高 8 位, 单位微秒
	0x0D	BW	7:4	0x98	W/R	带宽: 6: 62.5 kHz 7: 125 kHz 8: 250 kHz 9: 500 kHz
		CODING RATE	3:1		W/R	码率: 1: 4/5 2: 4/6 3: 4/7 4: 4/8
		Reserved	0		W/R	Reserved
	0x0E	SPREADING FACTOR	7:4	0x78	W/R	SF: 5: 32 chips / symbol 6: 64 chips / symbol 7: 128 chips / symbol 8: 256 chips / symbol 9: 512 chips / symbol 10: 1024 chips / symbol 11: 2048 chips / symbol 12: 4096 chips / symbol
		RXPAYLOADCRC CON	3		W/R	0: CRC 关闭 1: CRC 打开
		Reserved	2:0		W/R	Reserved
	0x0F	SYNCWORD[7:4]	7:5	0x12	W/R	同步字高 4bit



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						SF5 时取值范围 0x1~0x 3 SF6 时取值范围 0x 1~0x 7 其余 SF 取值范围 0x 1-0x F
		SYNCWORD[3:0]	3:0		W/R	同步字低 4bit SF5 时取值范围 0x0~0x 3 SF6 时取值范围 0x 0~0x 7 其余 SF 取值范围 0x 0-0x F
	0x10	IF[7:0]	7:0	0xC0	W/R	中频配置，次高 8 位
	0x11	IF[11:8]	7:4	0x0F	W/R	中频配置，高 4 位
		Reserved	3:0		W/R	Reserved
	0x12	Reserved	7:4	0x06	W/R	Reserved
		LOWDATARATE	3		W/R	0: 正常模式 1: 低速率模式
		Reserved	2:0		W/R	Reserved
	0x13	PREAMBLE_LEN[7:0]	7:0	0x08	W/R	preamble 长度低 8 位
	0x14	PREAMBLE_LEN[15:8]	7:0	0x00	W/R	preamble 长度高 8 位
	0x1A	SPI3_EN_REG	7	0x03	W/R	3 线 SPI 选择寄存器 0: 不选择 3 线 SPI 模式; 1: 选择 3 线 SPI 模式
		REG_MOSI_PUEN	6		W/R	mosi 管脚的寄存器控制模式 puen 控制寄存器
		REG_CSN_PUEN	5		W/R	csn 管脚的寄存器控制模式 csn 控制寄存器
		REG_SCK_PUEN	4		W/R	sck 管脚的寄存器控制模式 sck 控制寄存器
		REG_IN_PAD_MODE	3		W/R	SPI 输入管脚手动配置模式使能
		REG_OUT_PAD_MODE	2:0		W/R	SPI 的 MISO 管脚模式选择，用于 1 对多的 SPI 应用
	0x24	Reserved	7:4	0x00	W/R	Reserved
		EN_DCDC_3V	3		W/R	DCDC 使能 0: DCDC bypass 1: DCDC 工作
		Reserved	2:0		W/R	Reserved
	0x26	XTAL_ACTIVE_EN_3V	7	0x00	W/R	支持 3V 供电的有源晶振通道使能，保证芯片可工作在晶振 3V 时钟下 1: 打开；0: 关闭
		XTAL_STARTUP_EN_3V	6		W/R	晶振快速启动使能。1: 打开；0: 关闭。
		EN_DIG_COREPOWER_3V	5		W/R	数字核电压使能开关 1: 打开；0: 关闭
		Reserved	4:0		W/R	Reserved



13 Reference Schematic

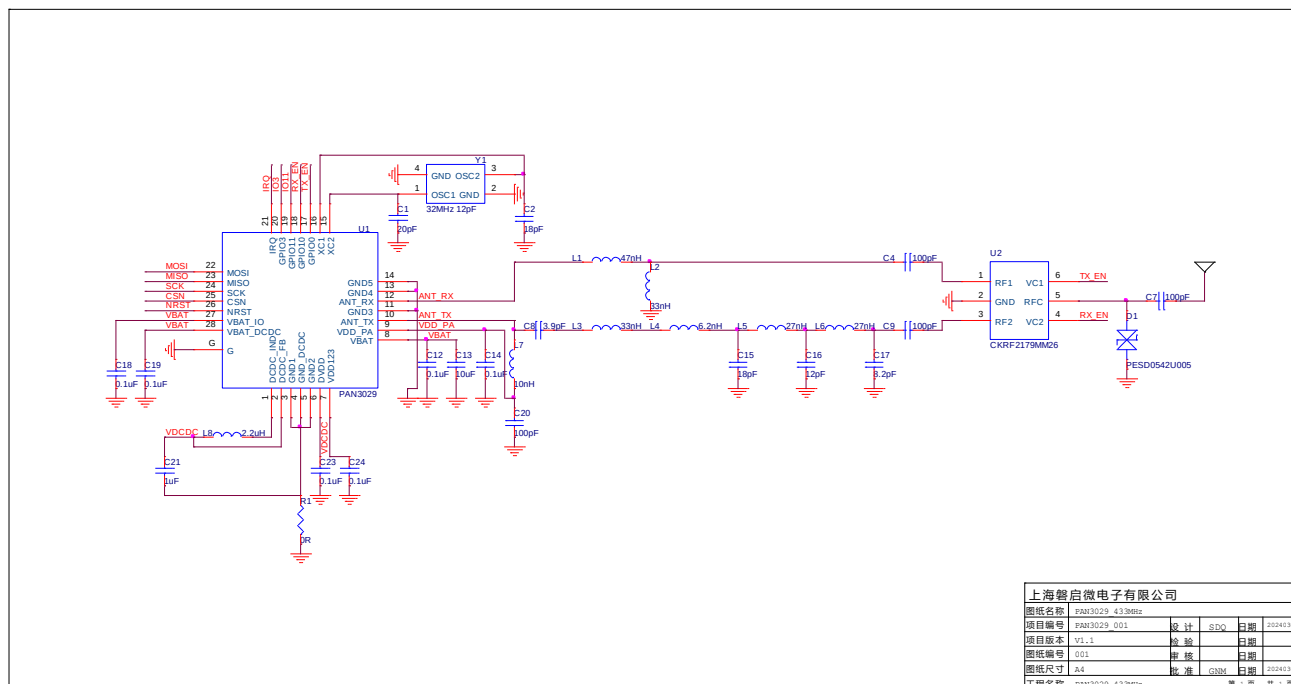


图 13-1 433MHz 参考原理图

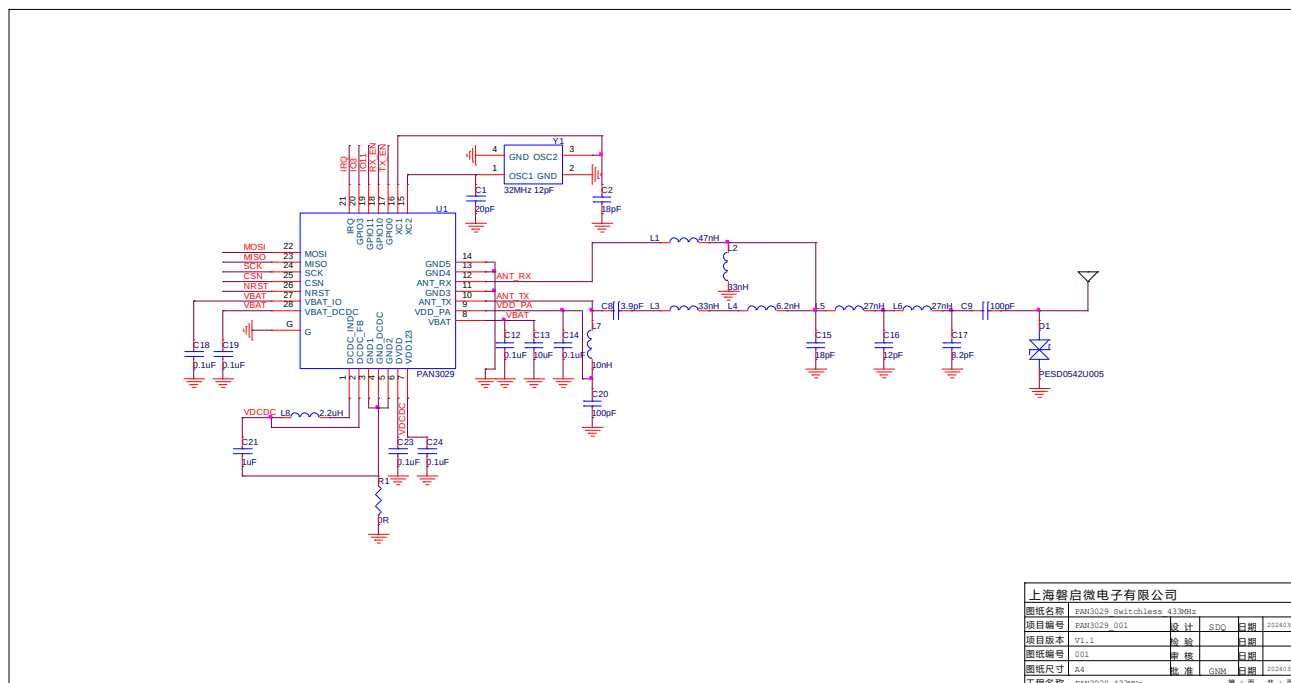


图 13-2 Switchless 433MHz 参考原理图



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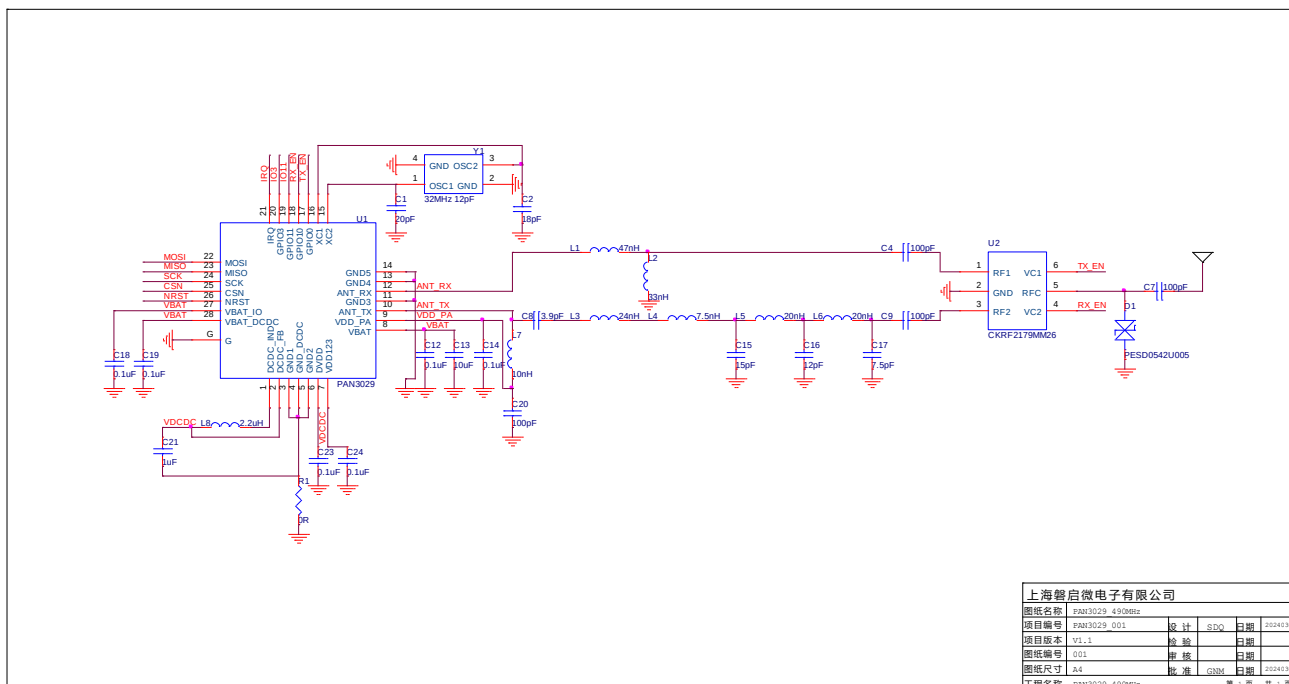


图 13-3 490MHz 参考原理图

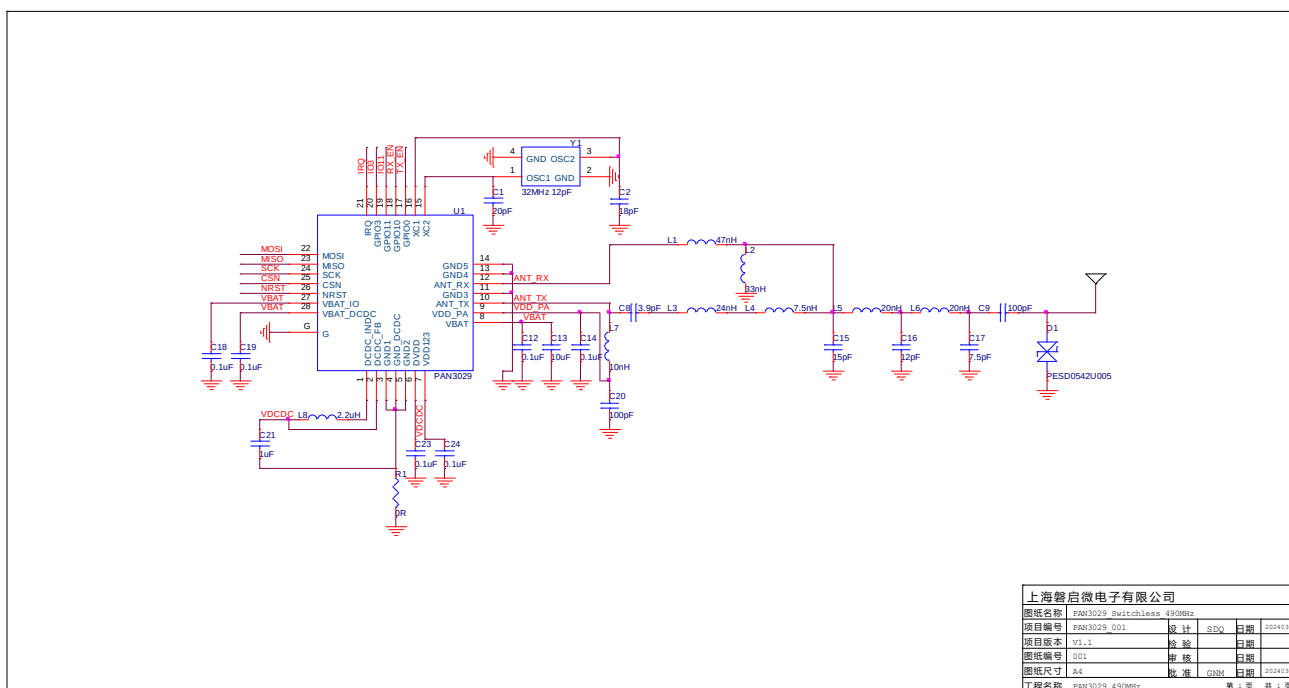


图 13-4 Switchless 490MHz 参考原理图



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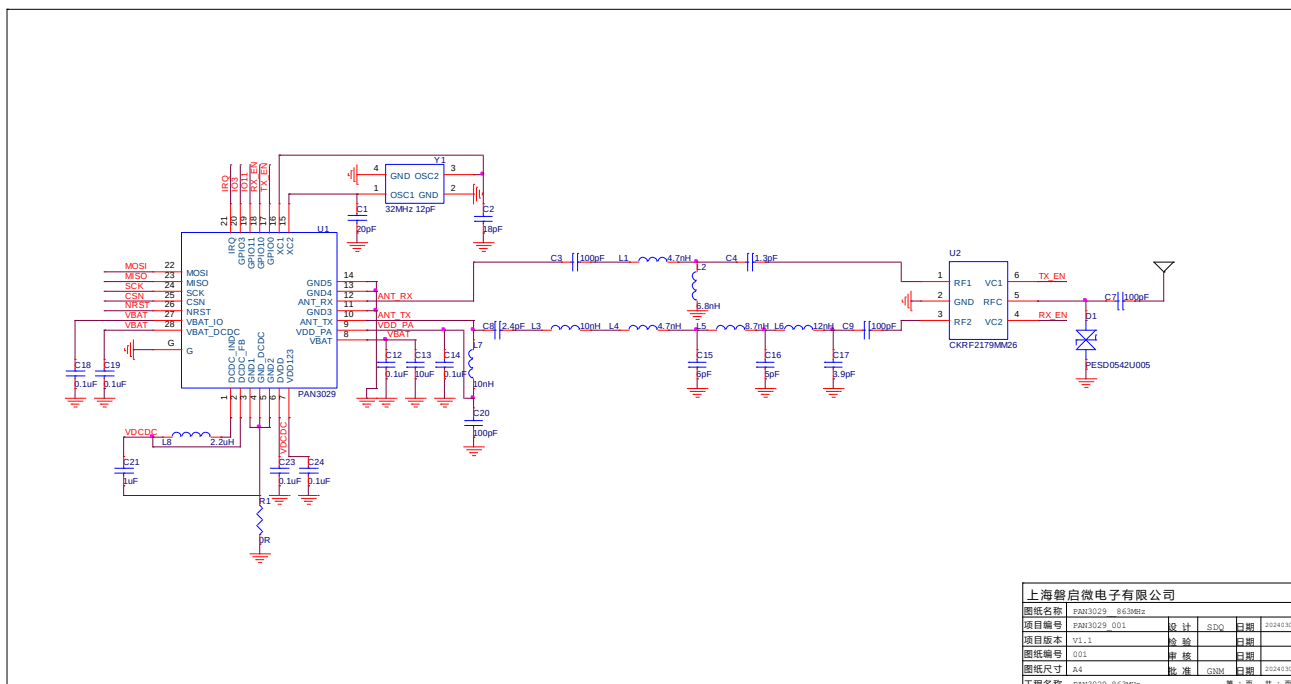


图 13-5 863MHz 参考原理图

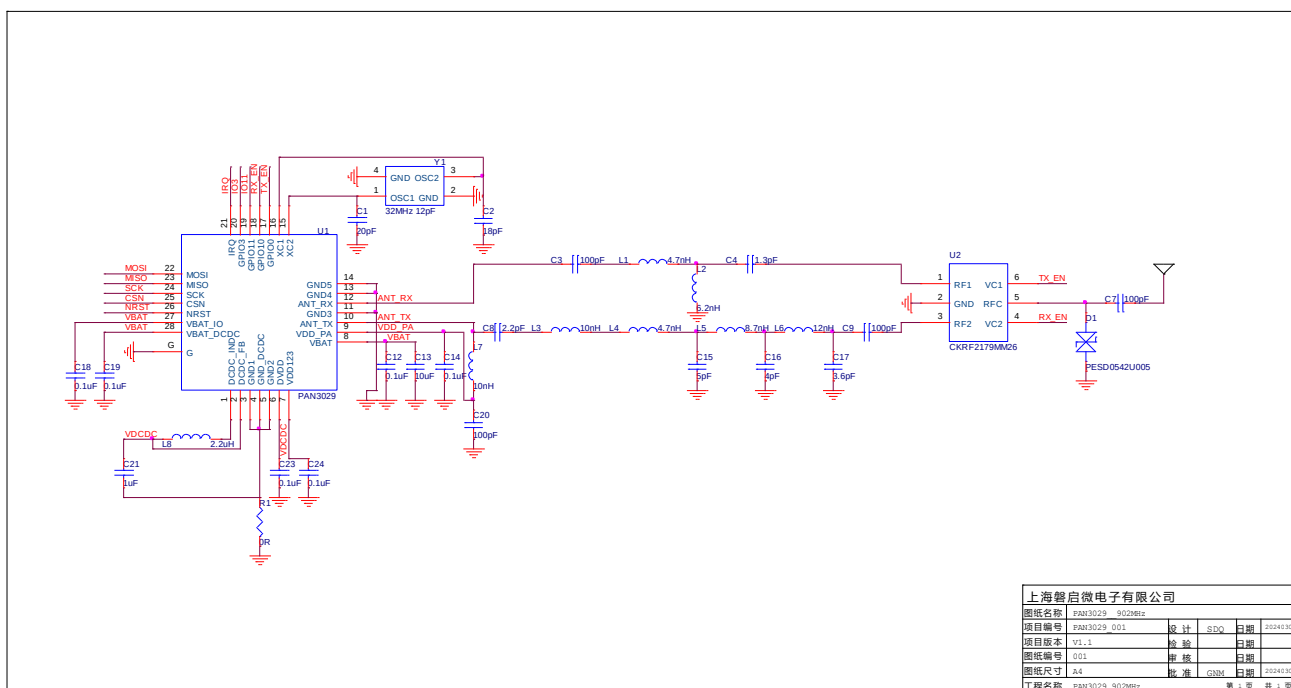


图 13-6 915MHz 参考原理图



14 Package size

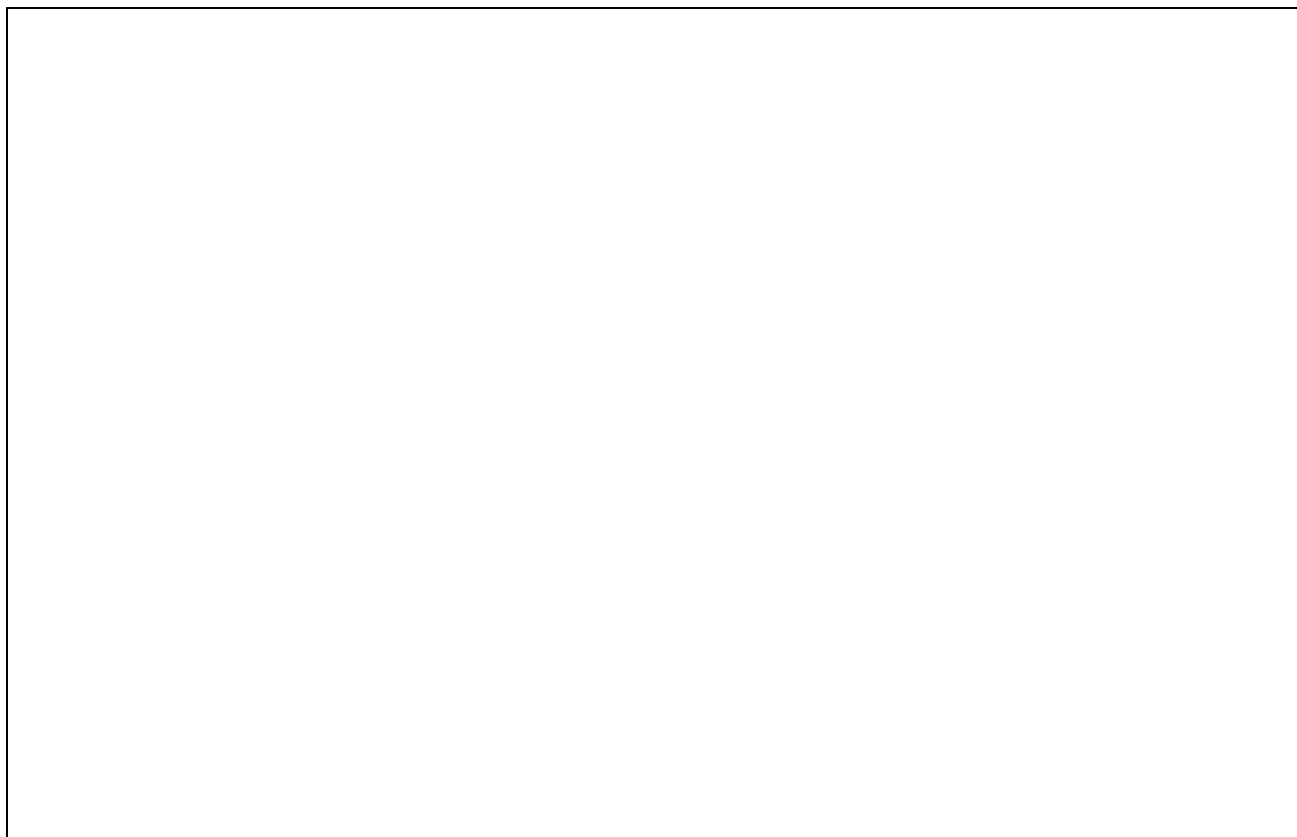


图 14-1 QFN28 封装图

表 14-1 QFN28 封装尺寸

Dimension	Min (mm)	Typical (mm)	Max (mm)
A	0.70	0.75	0.80
A1	-	0.02	0.05
A2	0.203 REF		
b	0.15	0.20	0.25
D	3.90	4.00	4.10
D2	2.55	2.65	2.75
E	3.90	4.00	4.10
E2	2.55	2.65	2.75
e	0.40 BSC		
K	0.225	0.275	0.325
L	0.35	0.40	0.45
h	0.30	0.35	0.40
Ne	2.40 BSC		
Nd	2.40 BSC		



15 Precautions

- 1) 该产品属 CMOS 器件，在储存、运输、使用过程中要注意防静电。
- 2) 器件使用时接地要良好。
- 3) 回流焊温度不能超过 260℃。

无铅回流焊工艺曲线如下图：

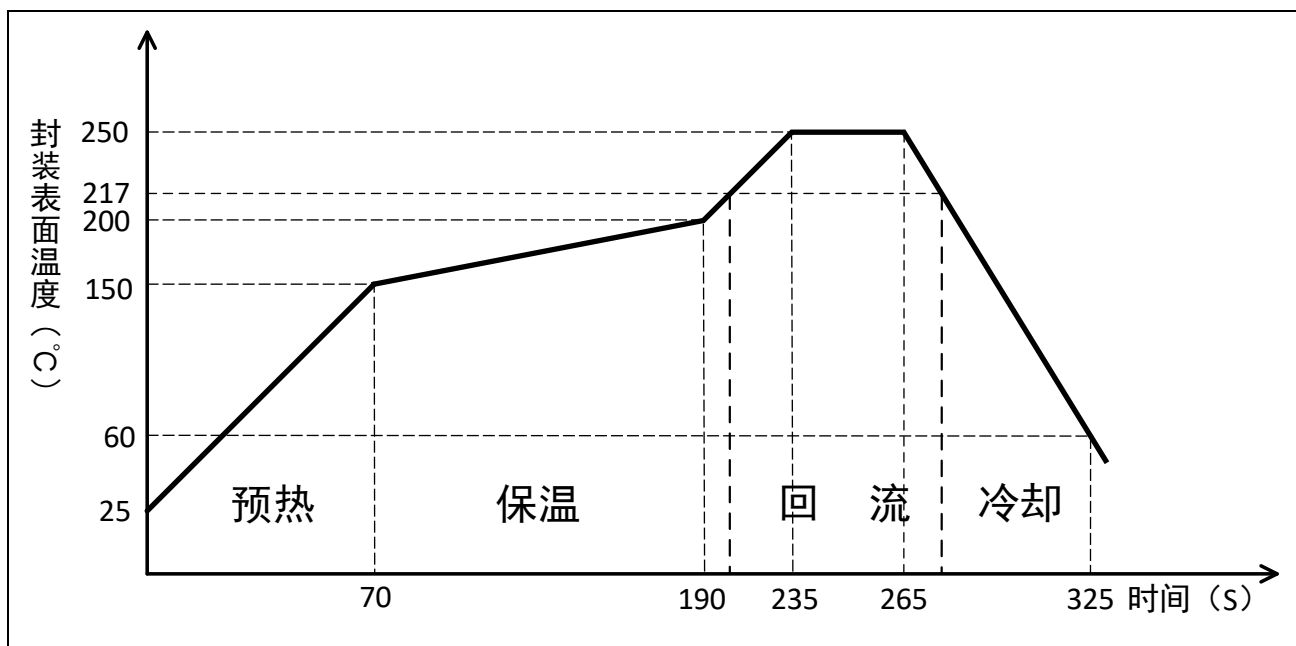


图 15-1 回流焊工艺曲线图



16 Storage conditions

- 1) 产品在密封包装中储存：在温度小于 30°C 且湿度小于 90%时，可达 12 个月。
- 2) 包装袋被打开后，元器件将被回流焊制程或其他的高温制程所采用时必须符合：
 - a) 在 72 小时内且工厂环境为小于 30°C≤60%RH 完成；
 - b) 保存在 10%RH 环境下；
 - c) 使用前进行 125°C，24h 烘烤去除内部水汽。
- 3) MSL（包装湿度敏感性）：3 级（根据 IPC/JEDEC J-STD-020 确定）



Abbreviations

ADC	模数转换器
CAD	信道活跃检测
Chirp	线性调频
CRC	循环冗余校验
CSN	SPI 片选信号
DAC	数模转换器
DCDC	直流变换器
FIFO	先入先出
GPIO	通用型输入输出
IRQ	中断请求
LDO	低压差线性稳压器
LPF	低通滤波器
MAC	介质访问控制层
MCU	微处理单元
Mixer	混频器
Modem	调制解调器
OSC	振荡器
PA	功率放大器
RF	射频
PLL	锁相环
PMU	电源管理单元
POR	上电复位
RAM	随机存取存储器
RSSI	信号强度指示
SCK	SPI 时钟信号
SF	扩频因子
SPI	串行外设接口
STB	待机模式
Sync	同步
VCO	压控振荡器



Revision History

Version	Date	Content
V1.0	2023.05	初版
V1.1	2023.06	增加包装方式、参考原理图，优化引脚定义中 pin1 和 pin2 名称，功能未变
V1.2	2023.07	增加 PAN3060 系列、电气特性等
V1.3	2023.08	更新参考原理图，补充工作电压描述
V1.4	2023.09	补充不同频段的参考原理图，补充不同频段的电气特性
V1.5	2023.10	更新参考原理图
V1.6	2023.12	更新基本操作配置中 SYNCWORD 寄存器的描述，更新参考原理图
V1.7	2024.03	更新参考原理图和电气特性。

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